BANK OF THAILAND

MONETARY POLICY REPORT

JUNE 2017

75th

Bank of Thailand
(1942-2017)

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
June 2017

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
Mr. Porametee Vimolsiri	Member
Mr. Apichai Boontherawara	Member
Mr. Sethaput Suthiwart-Narueput	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for the medium term and for 2017. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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Executive Summary

1. Global Economy

The global economy was projected to continue expanding. The U.S. economic growth outlook remained positive as consumption rebounded after a slowdown in the first quarter of the year due to temporary factors. Euro area economies continued to recover. Domestic demand was underpinned by accommodative monetary policy, improved labor market conditions, and reduced political uncertainties after a pro-EU political party claimed victory in the French presidential election; the recovery was also driven by improvements in exports thanks to global trade recovery. Meanwhile, the Japanese economy was recovering at a gradual pace on the back of a better export outlook. China continued to expand in the first quarter thanks to exports but was expected to slow down as a result of the ongoing economic structural reforms. Asian economies other than Japan and China continued to recover on the back of exports, especially electronics and commodities, which were driven by global demand. **Trading partners' headline inflation was expected to edge up slowly** while core inflation remained broadly stable. Overall, the **growth forecast for Thailand's trading partners was revised up. However, risks on the external front still tilted to the downside** in the medium term because of uncertainties surrounding U.S. economic and foreign trade policies, China's economic structural reforms, and global geopolitical risks.

Most central banks kept their policy rates on hold given the sluggish inflation outlook. On the other hand, the Federal Reserve raised the federal funds rate and announced its plans for balance sheet normalization following the Federal Open Market Committee (FOMC) meeting in June. In the period ahead, volatility in the global financial markets would likely remain high. Volatile capital flows, both into and out of emerging markets, might result from monetary policy actions of major advanced economies and uncertainties surrounding the U.S. economic stimulus policies.

Crude oil prices declined due mainly to supply-side factors, namely the increased oil output from U.S., Libyan, and Nigerian producers. Going forward, oil prices would slowly rise in tandem with the global economic recovery and production cuts by major oil producers. The Committee therefore revised the projection for the Dubai crude oil price down to an average of 50.9 U.S. dollars per barrel in 2017 before gradually edging up to 52.8 U.S. dollars per barrel in 2018.

2. Recent Economic Developments

The Thai economy continued to expand in the first quarter of 2017. Private consumption grew gradually, despite strong expansion of merchandise exports in a broader set of products. Government spending and tourism sector continued to be important growth drivers. Meanwhile, private investment contracted. However, **headline inflation slowed down in the second quarter** primarily due to supply-side factors. Particularly, fresh food prices sharply declined as a result of last year's high base following the drought coupled with this year's higher agricultural output thanks to favorable weather conditions. Energy prices also fell as a result of lower global oil prices.

Financial conditions remained accommodative. In the second quarter of 2017, Thai government bond yields of all, especially longer, tenures declined to a level below those at the end of last year. Financing costs through commercial banks decreased following cuts in commercial bank benchmark retail lending rates. Meanwhile, business financing through credit and bond markets continued to expand. Commercial bank loans picked up in the first two months of the second quarter mainly from business loans extended to large corporates, while loans to households rose across most loan purposes except credit card loans. With regard to exchange rates, the baht against the U.S. dollar appreciated in line with regional currencies due to the weakening of the U.S. dollar and increases in capital inflows into emerging markets including Thailand.

Financial stability remained sound overall, but there remained pockets of risks that warranted monitoring. The ratio of household debt to GDP trended down. However, the uneven economic recovery and structural problems faced by some business sectors undermined debt serviceability of both households and businesses, particularly SMEs. However, financial institutions maintained high levels of capital buffers and loan loss provisions to cushion against risks stemming from deteriorating credit quality. Regarding the real estate sector, a sharp increase in condominium prices as seen in earlier periods appeared less likely. Property developers and financial institutions were more cautious given slowing demand and large excess supply in some areas. Nevertheless, the existing level of real estate supply is deemed manageable by the property developers. Meanwhile, the search-for-yield behavior in the prolonged low interest rate environment remained present and might lead investors to underprice risks.

3. Economic Outlook

Looking ahead, Thailand's economic growth was projected to gain further traction and record 3.5 percent in 2017 and 3.7 percent in 2018. This upward revision to the previous forecast was mainly on account of the

pickup in merchandise and services exports. Meanwhile, private sector domestic demand continued to expand at a gradual pace.

The expansion in merchandise exports was more broad-based across product categories and export destinations. The value of merchandise exports in 2017 was projected to grow by 5 percent, an upgrade from last year's 0.1 percent growth. Export growth was expected to be driven particularly by exports of electronics that benefited from the increasing popularity of smartphones and Internet of Things (IoT) devices. Nonetheless, electronics export growth might soften somewhat in the period ahead as China's production of electronics started to slow down following prior acceleration to support the introduction of new smartphone models last year.

Exports of services recovered swiftly and were expected to continue expanding. The Committee revised up the projection for the number of foreign tourists in 2017 to 34.9 million, as the number of Chinese group tourists recovered swiftly from the impact of the measures to curb illegal tours toward the end of last year. Moreover, Thai tourism also benefited from the opening of new direct flight routes from China to major tourist destinations in Thailand and also from the Chinese government's restrictions on package tours to South Korea which diverted a portion of Chinese tourists to Thailand.

Public expenditure remained a key economic driver. Budget disbursement for both public consumption and public investment was well on track despite some unexpected delay for certain projects such as provincial development, new motorways, dual-track rails, and some new mass-transit lines under the government's infrastructure investment plan.

Private consumption expanded at a gradual pace, supported by improvements in farm income, together with employment in services and export-related manufacturing sectors that was expected to gradually increase. However, overall purchasing power had yet to fully recover as both employment and household income in the manufacturing sector did not benefit much from the export recovery, while the use of automation in place of manual labor was on the rise. In addition, elevated household debt continued to cause consumers to continue their cautious spending pattern, especially among lower income households with weaker debt serviceability than other income groups.

Private investment was projected to slowly recover. In the near term, investment expenditure to expand production capacity was observed in export-oriented manufacturing sectors especially electronics. However, outlays for construction trended down in tandem with the slowdown in the property market. In the period ahead, growth in consumption and exports, together with continued public investment, were expected to support the recovery in private investment.

Headline inflation remained low due to supply-side factors but was expected to gradually rise in tandem with demand-pull inflationary pressures that were expected to slowly increase given the improved growth outlook. However, inflation was still held down by oil prices, which were projected to remain largely unchanged or increase slowly, and also by sluggish increases in fresh food prices. The latter was attributable to (1) cyclical factors such as commodity food prices which stabilized at low levels and (2) structural factors such as the development of new manufacturing production and irrigation technology. **The Committee therefore revised down its forecasts for headline inflation to 0.8 and 1.6 percent in 2017 and 2018, respectively.**

The balance of risks to growth remained tilted to the downside and was largely unchanged from the previous assessment. While growth of overall trading partners' economies and global trade improved, risks on the external front remained. These included uncertainties pertaining to the U.S. foreign trade policy and geopolitical risks, both of which could have adverse economic impacts on Thailand's trading partners. However, there were still some possibilities that the Thai economy would achieve higher growth than the baseline projection if the U.S. economy were to exhibit stronger-than-expected growth due to stimulus measures. In contrast to the balance of risks to growth, the **balance of risks to inflation tilted to the upside** as more stringent regulations on immigrant workers could further tighten labor market conditions and in turn raise labor costs for businesses. The extent of the upside risk would depend on the implementation and management by the authorities.

4. Monetary Policy Deliberation

In the Monetary Policy Committee meetings on 24 May and 5 July 2017, the Committee unanimously voted to maintain the policy interest rate at 1.50 percent. In deliberating their policy decision, the Committee viewed that Thailand's economic growth continued to gain further traction driven by the external sector. Private consumption was expected to recover at a gradual pace. Meanwhile, headline inflation this year was projected to be lower than the previous assessment mainly due to supply-side factors. Nevertheless, inflation still retained an upward trajectory overall and was projected to rise in tandem with economic growth and domestic demand recovery.

Financial conditions remained accommodative and helped support economic growth overall. Financing costs decreased after a number of commercial banks cut their benchmark retail interest rates. At the same time, declines in bond yields also contributed to more accommodative financial conditions. Meanwhile, the baht's appreciation over the recent period was in line with movements of other regional currencies, and the Committee would to continue to monitor short-term capital flows going forward.

Financial stability remained sound. However, there remained pockets of risks that warranted monitoring such as the debt serviceability of SMEs that had yet to improve which in part reflected competitiveness issues. Meanwhile, the search-for-yield behavior in the prolonged period of low interest rates remained present as reflected by a continued increase in asset sizes of foreign investment funds (FIFs). Nonetheless, risks pertaining to FIFs were still limited as their investments were mostly in the form of deposits or short-term bonds in countries with good credit rating. In addition, investment in unrated bonds subsided following earlier defaults by some issuers, but the bond market overall underwent orderly adjustments. Going forward, the Committee viewed that close collaboration among regulatory agencies would be a key mechanism for effective monitoring, assessment, and management of various risks in the increasingly interconnected financial system, as it would reduce regulatory gaps and allow risks to be addressed in a more efficient and timely manner.

Monetary policy should remain accommodative to support the continuation of economic growth and to facilitate the return of headline inflation toward the target. While Thailand's growth outlook improved, the expansion of domestic demand was not yet sufficiently broad-based and the economy was still faced with risks on the external front. Inflationary pressures remained low and headline inflation this year might average below the target despite retaining an upward trajectory. Meanwhile, financial conditions eased and financial stability remained sound. In this light, the Committee concurred that the degree of monetary policy accommodation should be maintained for some time so that economic growth could continue and become more broad-based, which would enable headline inflation to return to target in a sustainable manner. The Committee would stand ready to utilize an appropriate mix of available policy tools to achieve these policy goals while also ensuring financial stability.

1. The Global Economy

Advanced economies continued to expand on the back of domestic demand recovery.

The U.S. economy continued to gain further traction. Domestic consumption growth resumed after a slowdown in the first quarter of 2017, due to temporary factors such as delayed tax refund, and was underpinned by improving macroeconomic fundamentals including the labor market that was approaching full employment and stronger household financial positions. Nonetheless, delays in the implemention of government stimulus measures could dampen business and household confidence going forward. The Committee also assessed the chances of the U.S. government passing a new budget to fund additional stimulus measures next year to be less likely, and therefore revised down the growth forecast for the U.S. economy to 2.1% in 2017. **Euro area economies exhibited a stronger recovery than previously assessed.** Domestic demand was supported by accommodative monetary policy, rising employment and decreased political uncertainties after the victory of Emmanuel Macron, a pro-EU candidate, in the French presidential election. Rising exports from the global trade recovery also contributed to manufacturing growth as reflected by continued increases in the Purchasing Manager's Index (PMI). **The Japanese economy recovered at a gradual pace,** driven mainly by exports growth which were underpinned by improving demand from Japan's trading partners. At the same time, monetary policy remained accommodative to support domestic consumption.

Nevertheless, there remained risks to the recovery of major advanced economies over the medium term from (1) uncertainties pertaining to the U.S. economic stimulus policies including tax reforms and infrastructure investments, together with the monetary policy direction of the Federal Reserve (Fed); (2) risks in the euro area which, despite a steady improvement overall, continued to warrant close monitoring. These risks include problems faced by European banks especially in Italy where non-performing loans remained problematic, political uncertainties especially the German election in the second half of the year, and Brexit negotiations between the UK and EU on trade and other agreements.

The Chinese economy slowed down due to ongoing economic structural reforms, while other Asian economies recovered on the back of exports, especially of electronics.

China's growth in the first quarter of 2017 picked up from the previous quarter on account of the recovery in exports and investment, together with government stimulus measures deployed to maintain growth. Based on the strong growth seen in the first quarter, the Committee revised up the growth forecast for the Chinese economy in 2017 from 6.5% to 6.6%. However, China would likely slow down due to ongoing economic structural reforms that placed greater emphasis on long-term economic stability. In addition, financial stability concerns in China continued to warrant close monitoring especially those regarding high corporate debt levels, speculation in the real estate sector, and capital outflows that remained a significant risk despite having already subsided somewhat. Against this backdrop, Moody's downgraded China's credit rating[1/] partly because of concerns over rising debt. Nevertheless, the impact of the China's credit rating downgrade on financial markets was

[1] From Aa3 to A1 and changed the rating outlook from negative to stable on 24 May 2017

limited given that concerns posed by Moody's were mostly priced in by the market. Moreover, China's external debt remained low. Hence, China's cost of borrowing did not increase significantly as a result.

Asian economies (excluding Japan and China) recovered at a gradual pace. Growth momentum from the recovery in the first quarter of 2017 was still intact, especially exports of electronics and commodities to China, Japan, and Europe. The expansion of Asian exports to China was partly due to investment stimulus efforts from the Chinese government as well as improvements in China's own exports which led to strong import demand in the first quarter of the year (Chart 1.1). Such developments reflected that Asian exports benefited from improvements in trading partners' demand and increases in commodity prices. Looking ahead, Asian exports were expected to continue being driven by the recovery in major advanced economies and growth of technological goods, especially Internet of Things (IoT) products (Chart 1.2), although they could be negatively affected by China's economic slowdown. In addition, there remained several risks facing Asian economies, including (1) exports-related risks from uncertainties in the U.S. foreign trade policies and shifts in the global trade structure whereby advanced economies and China now relied more on domestic production, together with relocation of production sites to regions with lower labor costs than Asia, and (2) risks to consumption from high levels of household debt in some countries, and possible tightening of liquidity conditions as a result from monetary policy normalization in major economies.

Chart 1.1 Asian exports increased in the first quarter from exports to China, Japan, and EU markets.

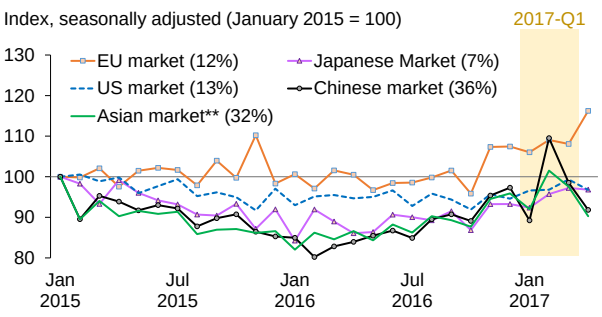
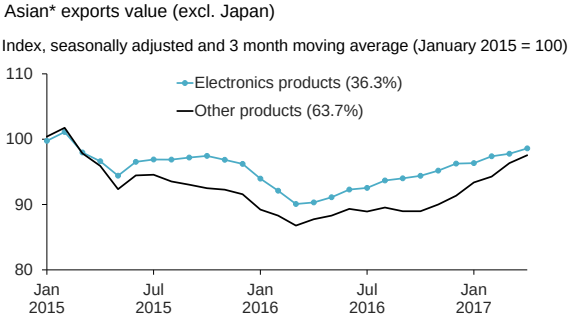


Chart 1.2 Asian exports were driven by exports of electronics and technological goods.



Note: Including China, Hong Kong, Taiwan, Korea, Malaysia, Singapore, and Thaiand

Source: CEIC

The growth forecast for Thailand's trading partners was revised up, but risks remained tilted to the downside overall.

The Committee revised up the growth forecast for Thailand's trading partners in 2017 from 3.3% to 3.4%. The growth forecast for 2018 was kept unchanged at 3.3% despite a downgrade to U.S. economic growth projection because growth momentum of other trading partners, especially in Asia, remained intact. Meanwhile, external risks were mostly tilted to the downside from (1) uncertainty in the U.S. foreign trade policies that could affect exports of China and Asia, (2) financial stability concerns in China, and (3) geopolitical risks from conflicts in the Korean Peninsula and the Middle East. Upside risks declined as the implementation of U.S economic stimulus became less likely. Nevertheless, downside risks also decreased as the anti-EU sentiment receded after the results of France's presidential election.

Table 1.1 Assumption on trading partners' economic growth

Annual change (%YoY)	Weight (%)	2016*	2017	2018
United States	14.9	1.6	2.1 (2.3)	2.1 (2.4)
Euro area	10.0	1.7	1.8 (1.5)	1.5 (1.5)
Japan	13.6	1.0	1.2 (1.1)	1.0 (0.9)
China	15.7	6.7	6.6 (6.5)	6.2 (6.2)
Asia (excluding Japan and China)**	37.4	3.5	3.8 (3.6)	3.8 (3.8)
Total***	100	3.1	3.4 (3.3)	3.3 (3.3)

Note: * Outturn
** Weighted by share of Thailand's total exports to 7 trading partners in 2014: Singapore (6.5%), Hong Kong (7.9%), Malaysia (8%), Taiwan (2.5%), Indonesia (5.9%), South Korea (2.8%), and the Philippines (3.7%)
***Weighted by share of Thailand's total exports to major 13 trading partners in 2014 (including the United Kingdom and Australia)
( ) reported in Monetary Policy Report March 2017

Most central banks kept their policy rates unchanged.

Despite improvements seen in global economic outlook, headline inflation remained sluggish in many countries while core inflation also remained largely stable. This was partly due to a weaker link between economic activities and inflation than in the past, as well as the fact that many economies did not yet fully recover. As a result, most central banks continued to maintain an accommodative monetary policy stance. However, demand-pull inflationary pressures edged up in some economies, especially the U.S. which was approaching full employment. The Fed hence raised the target range for the federal funds rate by 0.25 percentage point to 1.00-1.25 percent at the meeting on 13-14 June 2017, and announced its plans and procedures in reducing the size of its balance sheet. It was expected that the Fed would commence its balance sheet reduction process and continue to hike the federal funds rate in the near future.

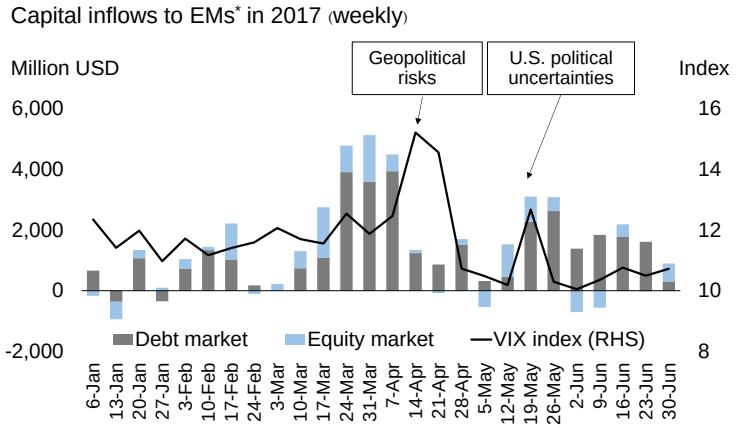
Capital flows into emerging markets slowed down in the first half of the second quarter of 2017 from geopolitical risks, but picked up later in the latter half of the quarter.

Toward the end of the first quarter of 2017, capital inflows to emerging markets (EMs) continued as market participants anticipated the Fed's policy rate hikes to be more gradual than previously expected. In mid-April, capital inflows slowed down due to heightened geopolitical risks in the Middle East and the Korean Peninsula (Chart 1.3) as reflected by a spike in the VIX index.[2/] The VIX index rose again in mid-May as news broke out about President Donald Trump's possible impeachment for obstruction of justice and for firing the FBI director. At the same time, key U.S. economic indicators such as those concerning production, labor market conditions, and inflation turned out softer than the market previously expected. Investors thus revised down the likelihood of the Fed's rate hikes in the second half of the year, and redirected their attention toward EM assets. This shift resulted in increases in capital inflows to EMs especially into bond markets. Looking ahead, global financial markets would likely remain volatile. Abrupt capital flows movement, both into and out of EMs, might

[2] VIX index indicates volatility of stock markets, calculated by implied volatility from the S&P 500 options.

also result from the conduct of monetary policy in major advanced economies, uncertainties pertaining to the U.S. economic stimulus, concerns over the European and Chinese financial sectors, and geopolitical risks.

Chart 1.3 Capital inflows to EMs slowed due to geopolitical risks, but resumed in the second half of Q2-2017.



Note: *EMs includes Thailand, Indonesia, South Africa, and Turkey
Sources: Bloomberg and International Institute of Finance

The Dubai crude oil price declined in the second quarter of 2017 mainly due to supply-side factors, but was expected to slowly trend up given the global economic recovery and production cuts by oil producers.

The Dubai crude oil price in the second half of 2017 declined from the previous quarter mainly due to supply-side factors in the U.S. Even though OPEC and non-OPEC oil producers had cut their production since the beginning of the year, oil prices continued to face downward pressures from several supply-side factors. In particular, these factors were (1) the larger-than-expected oil output from U.S. producers as the number of oil rigs and productivity per rig at these new production sites continued to increase, and (2) the resumption of oil production in Libya and Nigeria after domestic unrests subsided and both countries were exempt from OPEC's agreement on production cuts.

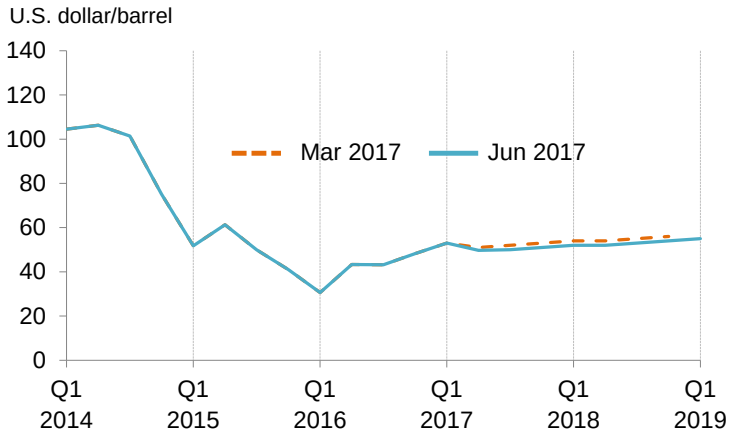
Oil prices were projected to gradually increase in the period ahead, albeit at a slow pace (Chart 1.4). Key supporting factors include growing oil demand from the ongoing global economic recovery, together with the production cut agreement between OPEC and non-OPEC producers that was extended for another 9 months until the first quarter of 2018. Nonetheless, the large stock of global crude oil and the expansion of shale oil production in the U.S. would still continue to restrain increases in oil prices going forward.

Against this backdrop, the Committee revised down the projection for Dubai oil price from 52.3 to 50.9 U.S. dollars per barrel in 2017, and from 54.8 to 52.8 U.S. dollars per barrel in 2018. The balance of risks to the projection was more balanced compared with the previous assessment where it was tilted to the downside. This adjustment to the balance of risks was due to the greater likelihood that OPEC and non-OPEC producers would comply with their agreement to cut production and might add new measures to prop up oil prices. At the same time, intensified geopolitical tensions could push up oil prices in some periods. Downside risks to the oil price projection remained, however, as global crude output might turn out higher than

expected especially from (1) shale oil production in the U.S. and (2) the resumption of oil production in Libya and Nigeria whose output might exceed expectations.

Chart 1.4 Dubai oil price declined in second quarter but was expected to slowly trend up in the period ahead.

Assumption on Dubai oil price



Metal prices decreased slightly in the short term, mainly as a result of (1) larger-than-expected global stock of iron ore due to increased supply and (2) a slowdown in demand for iron ore. On the whole, metal prices were nevertheless projected to edge up in 2017 on account of production cuts in China's four major provinces and the ongoing infrastructure investment in China. Metal prices in 2018 were projected to remain mostly unchanged from the previous year, similar to the previous assessment. In addition, metal prices might face downward pressure from supply-side factors namely the increased production from Chinese producers who had cut production earlier.

2. The Thai Economy

2.1 Recent developments

The Thai economy continued to expand in the first quarter of 2017. Private consumption grew gradually, despite strong expansion of merchandise exports in a broader range of products. Public spending and tourism continued to be important growth drivers. Meanwhile, private investment contracted.

The Thai economy expanded 3.3 percent in the **first quarter of 2017** from the same period last year. Growth was primarily driven by private consumption, especially of durable goods, due to improved consumer confidence and a continued recovery in farm income. Growth in merchandise exports was more broad-based across product categories and major export destinations. Exports of services recovered swiftly from the impact of measures to curb illegal tour operators at the end of last year. Meanwhile, public spending continued to grow in tandem with robust investment expenditure from both the government and state-owned enterprises. This was especially the case for ongoing state-owned enterprise projects such as the CAT Telecom's internet networks and the Provincial Electricity Authority's power distribution system. These factors helped offset the contraction in private investment which to some extent reflected the production capacity that was not fully utilized. Overall, Thailand's economic growth accelerated 1.1 percent in the first quarter of 2017 after seasonal adjustments, an improvement from 0.6 percent in the previous quarter, with growth momentum led by a sustained recovery in external demand.

Thailand's economic growth continued to gain traction over the **first two months of the second quarter**. Private consumption slowed down following prior acceleration of spending on durable goods in the previous quarter, but was still supported by improvements in consumer confidence and farm income. However, non-farm income had yet to fully benefit from the export recovery. With regard to exports, merchandise exports expanded both in terms of volume and price, and were more broad-based. This development was in line with the recovery of external demand and the increasing popularity of Internet of Things (IoT) products. Regarding tourism, growth improved across all tourist groups, especially Chinese nationals. This was partly a result of China's political conflicts with South Korea and Taiwan which diverted a portion of Chinese tourists to Thailand. Meanwhile, public spending remained a key growth driver despite slight disbursement delays for some construction projects of state-owned enterprises such as the red commuter rail line (Bang Sue - Rangsit) due to problems accessing construction sites, while some of mass transit lines and the second phase of the dual-track railway project were still under review and subject to contract amendments. Private investment remained low but started to show improvement signs as evidenced by increasing business confidence and investment expenditure in some export-related industries that had recovered such as electronics and electrical appliances.

Household purchasing power had yet to fully recover despite improvements seen in some household groups, partly because employment in the manufacturing sector did not benefit much from the export recovery coupled with the increasing adoption of automation in manufacturing processes.

Private consumption expanded at a gradual pace despite income increases among **agricultural households** from higher prices and output of major crops such as rice, fruits, and rubber. While such improvements in farm income were starting to become more widespread across different regions in Thailand, purchasing power did not increase correspondingly as debt levels remained elevated for some agricultural households. Purchasing power of **households outside the agricultural sector**, particularly those with low income, had yet to fully recover, partly because of high debt burden. In addition, employment in the manufacturing sector trended down, while employment in the services sector remained stable. This was because employment did not fully benefit from the recent recovery in exports and thus increases in employment were still limited to certain industries, namely food, electronics and fabricated metal. Average earnings in most manufacturing sectors also had yet to increase. As a result, overall purchasing power remained soft. In addition, employment in the manufacturing sector, especially in labor-intensive industries, faced additional pressures from the growing adoption of automation in manufacturing processes as means to respond to prior labor shortages and to enhance productivity.

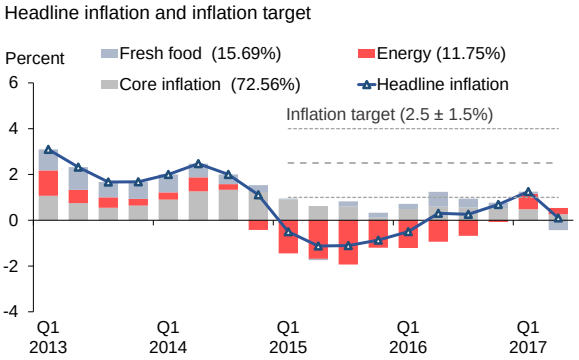
Looking ahead, more stringent regulations on immigrant workers[3] might tighten labor market conditions somewhat as a portion of immigrant workers returned to their home countries to undergo proper hiring processes required by law. This would affect certain sectors that were heavily reliant on immigrant workers such as construction, agriculture, labor-intensive industries, and trade sectors and might put pressures on raising labor costs. Nevertheless, the extent of the impact would depend on the implementation and management of the new regulations by the government. Therefore, the Committee would closely monitor the situation and the impacts of the tighter regulations going forward.

Headline inflation declined mainly on account of supply-side factors, while core inflation slowed down from the previous quarter.

Headline inflation declined from the previous quarter, averaging at 0.17 percent in the second quarter of 2017 (Chart 2.1). The decline in headline inflation was attributable to lower fresh food prices, notably of vegetables and fruits (Chart 2.2) due to (1) last year's high base following the drought and (2) this year's higher agricultural output thanks to favorable weather conditions. Meanwhile, retail oil prices also declined with global crude oil prices. The one-year-ahead inflation expectations of businesses were largely unchanged from the previous figure at 2.1 percent, while the 5-year ahead inflation expectations of professional forecasters declined slightly to 2.3 percent, which remained close to the midpoint of the target at 2.5 percent.

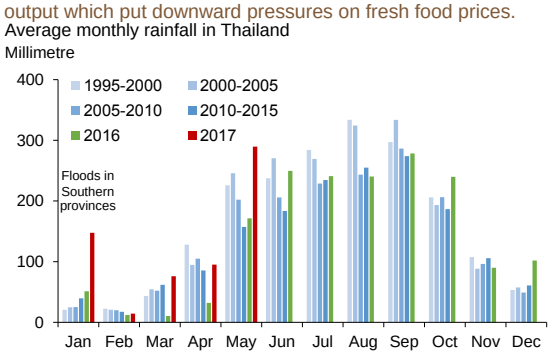
[3] The Royal Decree on Managing the Work of Aliens B.E. 2560 (2017) became effective on 23 June 2017. The new Royal Decree amended the previous legislations to cover the management of immigrant workers by merging the contents in the Alien's Work Act B.E. 2551 (2008) and the Royal Decree on Bringing Alien to Work in the Kingdom B.E. 2559 (2016). The new Royal Decree emphasized on protecting and assisting both Thai employers and immigrant workers, and increasing penalties for both employers and immigrant workers that engaged in illegal hiring. However, the government recently extended the reprieve period to 180 days.

Chart 2.1 Headline inflation declined from the previous quarter mainly due to fresh food prices.



Note: ( ) denotes share in inflation basket
Source: Ministry of Commerce, calculations by the Bank of Thailand

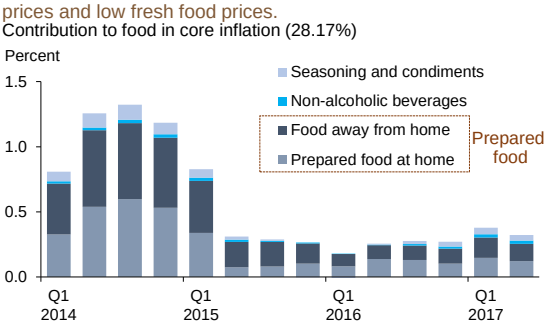
Chart 2.2 Favorable weather conditions, as reflected by higher than historical average rainfall, led to higher agricultural output which put downward pressures on fresh food prices.



Source: The Meteorological Department and the World Bank

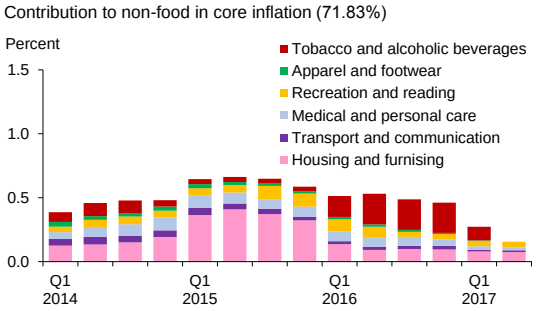
Core inflation softened from the first quarter, averaging at 0.47 percent in the second quarter of 2017. The main reasons behind the slowdown were: (1) the decrease in food prices, particularly processed food, on account of lower costs due to lower fresh food prices from last year's high base following the drought and this year's higher output as a result of favorable weather conditions (Chart 2.3) and (2) the decline in non-food prices following last year's higher base price of tobacco after the increase in excise tax on tobacco (Chart 2.4).

Chart 2.3 Food-in-core inflation remained low due to subdued cost-push inflationary pressures given the contraction in LPG prices and low fresh food prices.



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by the Bank of Thailand

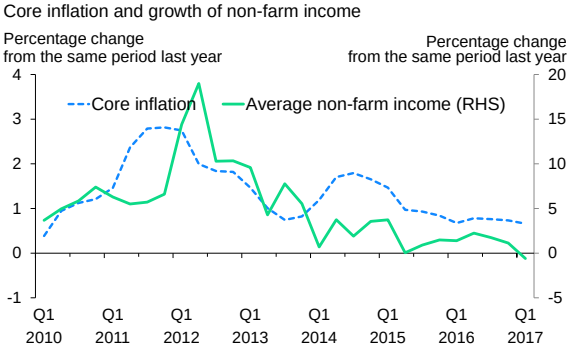
Chart 2.4 Non food-in-core inflation slowed down due to the base effect of last year's higher tobacco prices following an increase in excise tax on tobacco.



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by the Bank of Thailand

Overall, demand-pull inflationary pressures remained low as the recovery of household purchasing power was not yet broad-based. This was reflected by a continued slowdown in average non-farm income (Chart 2.5). Meanwhile, cost-push inflationary pressures were limited given the decline in oil prices and thus the cost-price pass-through was limited. Businesses also faced difficulties in raising product prices as reflected in the latest Business Sentiment Survey[4/] that revealed a higher ratio of respondents indicating greater difficulties in raising prices.

Chart 2.5 Average non-farm income saw subdued recovery, reflecting low demand-pull inflationary pressures.



Source: National Statistics Office and Ministry of Commerce

[4] The Business Sentiment Survey covers about 600 business operators per month across the country.

Table 2.1 Inflation

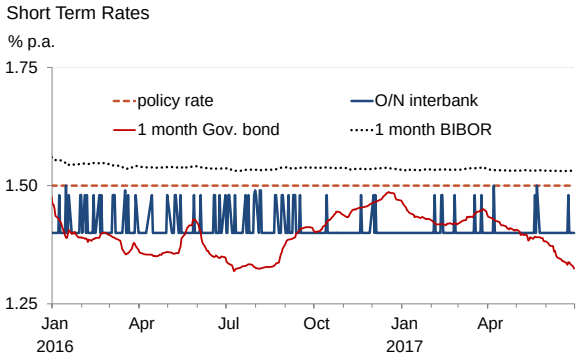
Annual percentage change	2015	2016				2017	
	Q4	Q1	Q2	Q3	Q4	Q1	Q2
Headline Consumer Price Index (Headline CPI)	-0.86	-0.50	0.30	0.26	0.69	1.25	0.10
Core Consumer Price Index (Core CPI)	0.85	0.67	0.79	0.76	0.73	0.66	0.47
Raw food	1.45	1.52	4.24	2.58	1.54	0.61	-2.97
Energy	-14.63	-11.41	-8.95	-7.00	-1.06	6.69	2.67

Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Government bond yields for all, especially longer, tenures declined.

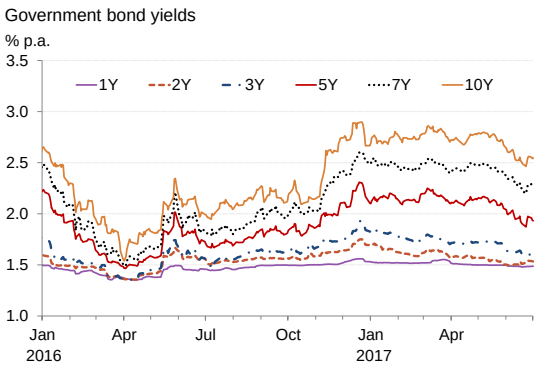
Short-term money market rates remained close to the policy rate in the second quarter of 2017. However, **short-term government bond yields** declined since the beginning of April (Chart 2.6) following the Bank of Thailand's announcement to reduce its issuances of short-term bonds to curb short-term baht speculation by foreign investors.[5/] Meanwhile, movements in **medium- and long-term government bond yields** were mainly influenced by external factors. Consequently, yields rose slightly in April following the release of the minutes of the Federal Open Market Committee (FOMC) meeting on 14-15 March 2017 which prompted market expectations that the Federal Reserve (Fed) would soon commence its balance sheet normalization, while investors were also concerned with the pending outcome of the French presidential election. In the latter half of May, lower-than-expected U.S. economic data outturns coupled with political concerns in the U.S. put downward pressures on U.S. government bond yields. In addition, although the Fed raised the federal funds rate again in June, the Fed also revised down its inflation forecast for 2017. Hence Thai government bond yields were driven down in tandem with U.S. government bond yields. Later in June, however, investors priced in a high probability that the European Central Bank (ECB) would reduce monetary policy accommodation after the ECB President expressed confidence over the growth outlook of the euro area and noted that inflationary pressures strengthened. As a result, Thai government bond yields rose slightly alongside increases in global bond yields. Overall, Thai government bond yields in this quarter declined to levels below those at the end of 2016 across all, particularly longer, tenures (Chart 2.7). **Corporate bond yields** also declined across the board alongside decreases in government bond yields, while credit spreads remained largely stable.

Chart 2.6 Short-term government bond yields trended down while other short term rates remained close to policy rate.



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Chart 2.7 Movements in medium- and long-term government bond yields were mainly driven by external factors.



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

[5] The Bank of Thailand reduced the issuance of 3-month and 6-month bonds since 5 April 2017 from 40 billion to 30 billion baht for each type.

Financing costs through commercial banks declined following cuts in commercial bank benchmark retail interest rates to aid retail borrowers in mid-May 2017. At the end of the second quarter, the minimum lending rate (MLR), minimum overdraft rate (MOR) and minimum retail rate (MRR) of large commercial banks decreased by 0.06, 0.12 and 0.44 percentage point from the previous quarter, respectively (Chart 2.8).

Chart 2.8 Commercial banks cut their benchmark retail interest rates from May to June 2017.

Commercial banks benchmark retail interest rates*

% p.a.	2016	Q1/17	May 17	Jun 17	4 Jul 17
MLR					
Average of 4 largest banks**	6.26	6.26	6.20	6.20	6.20
Average of other banks	7.16	7.16	7.16	7.15	7.15
MOR					
Average of 4 largest banks	7.18	7.18	7.06	7.06	7.06
Average of other banks	8.01	8.01	7.99	7.96	7.96
MRR					
Average of 4 largest banks	7.62	7.62	7.18	7.18	7.18
Average of other banks	8.45	8.45	8.29	8.24	8.24

Notes: *End of Period
**Bangkok Bank, Krung Thai Bank, Kasikorn Bank, and Siam Commercial Bank

Source: Bank of Thailand

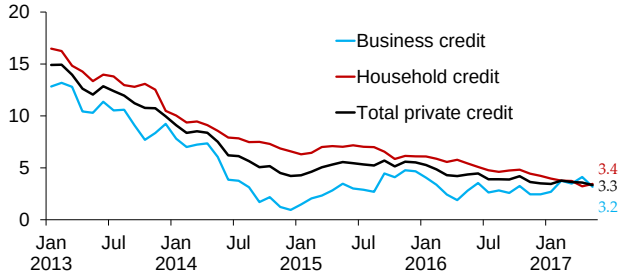
Private credit continued to rise

Private credit[6] expanded 3.6 percent from the same period last year in the first quarter of 2017. Commercial bank loans rose slightly mainly on account of business loans to large corporates. In May 2017, private credit grew 3.3 percent from the same period last year as household credit slowed down, partly as a result of last year's high base following the acceleration in mortgage loan applications before the real estate stimulus measures expired. Business credit softened in part due to a high base last year from loans to holding companies to finance their tender offers and offshore business acquisitions.

Chart 2.9 Growth in private credit improved.

Growth of private credit

Percentage change from the same period last year



Note: Outstanding credit of other depositary corporations (ODC) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds

Source: Bank of Thailand

[6] Outstanding credit of other depository corporations (ODCs) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives and money market mutual funds.

The change in business credit outstanding between April and May 2017 continued to edge up since the end of the first quarter especially for loans to large corporates. Credit growth was also more broad-based, having expanded across more business sectors such as hotels, real estate, food manufacturing, transportation, wholesale and retail trade sectors. As for other financing channels, **funding through the bond market** increased especially among businesses related to financial institutions, real estate, construction materials, energy and telecommunication. **Funding through the equity market** increased markedly in May where new shares were issued by construction material companies to finance investment abroad.

The change in household credit outstanding in April 2017 slowed relative to the end of the first quarter, but accelerated somewhat in May on account of credit extended by commercial banks and specialized financial institutions. In particular, growth in consumer credit extended by commercial banks accelerated in almost all loan purposes, except credit card loans. Meanwhile, growth in consumer credit extended by specialized financial institutions accelerated on the back of mortgage loans.

Going forward, financial conditions were expected to remain accommodative as many commercial banks had cut their benchmark retail interest rates in mid-May, and as Thai government bond yields declined due to external factors. However, it was expected that financial institutions would still maintain caution in extending credit especially to businesses with deteriorating credit quality.

The baht appreciated against the U.S. dollar in line with movements of regional currencies.

In the second quarter of 2017, the baht underwent volatile movements and recorded a net appreciation against the U.S**.** dollar relative to the end of the first quarter. Movements in the baht was in line with those of most regional currencies which also appreciated as the U.S. dollar weakened, particularly in April and May (Chart 2.10). In addition, rising geopolitical tensions in the Middle East and over the Korean Peninsula and political concerns the U.S., together with weaker-than-expected U.S. economic data outturns that led the market to expect a lower probability of another rate hike by the Fed in the latter half of this year, induced larger capital flows into yen-denominated assets and into emerging markets including Thailand.

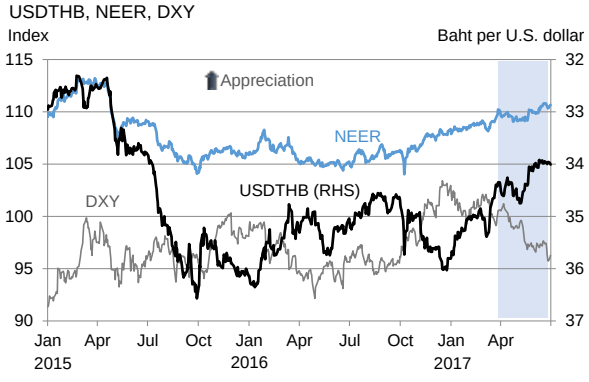
However, the Fed's decision to raise its federal funds rate and announcement of its plans for gradual balance sheet normalization at the FOMC meeting on 13-14 June 2017 were largely expected by the market, and thus did not stir up volatility in global financial markets. With regard to the baht's stronger appreciation compared with those of regional currencies during some periods in May, such extent of appreciation was primarily due to profit and dividend repatriation by foreign companies in Thailand and an acceleration in U.S. dollar purchases by some local gold dealers when gold prices fell. As of 4 July 2017, the baht appreciated 1 percent from the end of the first quarter, closing at 33.93 baht to the U.S. dollar.

The **nominal effective exchange rate** (**NEER**) stood at 110.65 on 4 July 2017. This was equivalent to a 0.4 percent appreciation from the previous quarter, driven mainly by the baht's appreciation against the Japanese yen, Korean won[7], and U.S. dollar. However, such appreciation was offset by the baht's depreciation against the euro following the outcome of

[7] Korean won weakened on concerns over geopolitical risks in the Korean Peninsula.

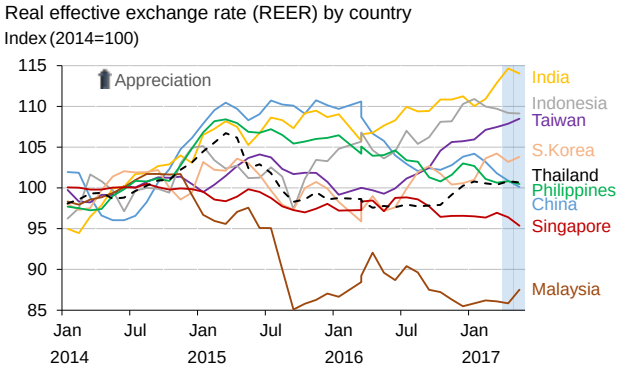
the French presidential election and the ECB President's optimistic remarks regarding the strength of the euro area's economic recovery and inflation outlook. The **real effective exchange rate (REER)** stood at 102.45 as of May 2017 which represented a 0.4 percent appreciation from March. Appreciation of Thailand's REER was moderate compared with regional peers (Chart 2.11). Overall, the REER strengthened 0.5 percent from December 2016 which was less than the NEER's appreciation of 1.5 percent. This reflected that Thailand's relatively low inflation compared with trading partners helped alleviate to some extent the impact of NEER's appreciation on price competitiveness. In the period ahead, exchange rates would likely remain volatile due to external uncertainties such as those related to the U.S. economic stimulus policies and monetary policy actions of major advanced economies.

Chart 2.10 The baht strengthened against the U.S. dollar as the U.S. dollar weakened across the board.



Source: Bank of Thailand and Reuters (data as of 4 July 2017)

Chart 2.11 Thailand's real effective exchange rate appreciated rather moderately in comparison to other regional economies.

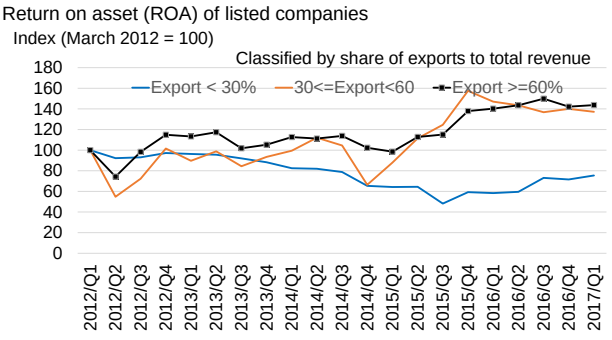


Source: BIS, calculations by the Bank of Thailand (data as of May 2017)

Financial stability remained sound but the uneven economic recovery and structural problems in some sectors gave rise to pockets of risks that warranted close monitoring, such as debt serviceability of households and SMEs, an oversupply of housing in certain areas and price ranges, and the search-for-yield behavior.

Thailand's financial stability remained sound. Financial institutions maintained high levels of capital buffers and loan loss provisions to cushion against risks stemming from deteriorating credit quality. Financial positions of listed companies improved overall, especially export-related businesses, as reflected in higher returns on assets (ROA) (Chart 2.12). Meanwhile, Thailand's external stability remained strong and able to provide cushion against volatilities in global financial markets with high level of foreign exchange reserves and sustained current account surpluses. Nevertheless, there were pockets of risks that warranted monitoring. For instance, households' debt serviceability remained fragile due to elevated debt levels. Even as the ratio of household debt to GDP gradually decreased since the first quarter of 2016, households would still require some time to adjust. Meanwhile, debt serviceability of SMEs continued to deteriorate, especially of those

Chart 2.12 Export businesses saw improvements to their balance sheet and financial positions.



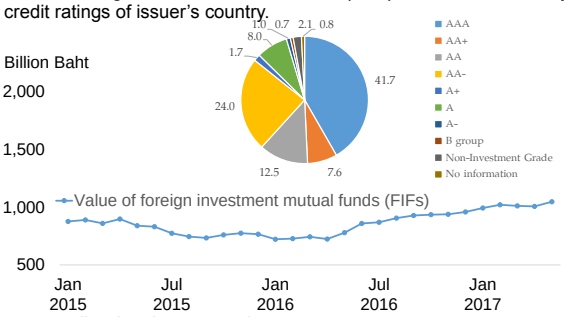
Sources: Stock Exchange of Thailand and Thai Customs, calculations by the Bank of Thailand

in commercial and manufacturing sectors, which reflected the uneven nature of the ongoing economic recovery and also structural problems faced by SMEs in adapting to the changing business environment. In the real estate sector, there remained an oversupply of condominiums in certain areas and price ranges, while demand for housing remained flat at low levels when compared with historical averages. Furthermore, the continued deterioration in the credit quality of housing loans might lead to more stringent credit standards from commercial banks which might in turn affect demand in the period ahead. (see Article: *The property market in Bangkok and vicinities*)

The search-for-yield behavior remained present in the prolonged low interest rate environment. Investment in mutual funds continued to grow, notably in foreign investment funds (FIF) which mainly invested in deposits and short-term debt securities in countries with investment-grade credit ratings (A Group) (Chart 2.13). Meanwhile, investment in funds with deposit-like characteristics picked up slightly in April after having decelerated in the latter half of 2016. In addition, investment in unrated bonds trended down following defaults at the end of 2016 as issuers of those defaulted bonds switched to issuing longer-term debts or borrowing from commercial banks instead. The situation on unrated bond rollovers improved overall and funding costs remained largely unchanged. In sum, adjustments in the bond market in response to unrated bond defaults were orderly so far. Nevertheless, debt serviceability on maturing unrated bonds, especially those of issuers with fragile financial positions, continued to warrant close monitoring going forward.

Chart 2.13 Most FIFs invested in countries with investment-grade credit ratings (A group).

Value of foreign investment mutual funds (FIFs) and share of FIFs by credit ratings of issuer's country.



Note: credit ratings by S&P as of 30 June 2017
Source: Association of Investment Management Companies (AIMC), calculations by the Bank of Thailand

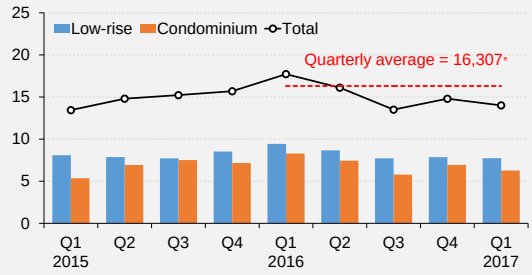
The property market in Bangkok and vicinities

The property market in Bangkok and vicinities slowed down in the first quarter of 2017. Housing **demand** weakened as household purchasing power was not sufficiently strong, coupled with lower loan affordability of households given prevailing high debt burden. Hence, commercial banks were more cautious in mortgage lending as reflected in the smaller number of new mortgage loan approvals in the first quarter of 2017 for both low-rise and condominium units, which fell from the previous quarter and stood below historical averages[8] **(Chart 1)**. On the **supply** side, developers deferred new project launches in line with a decline in demand at the beginning of the year. This was evident in the fewer number of new residential projects launched as compared with the previous quarter **(Chart 2)**.

Chart 1 Housing demand was below historical averages.

Residential units in Bangkok and vicinities with approved mortgages by commercial banks

Thousand units, seasonally adjusted



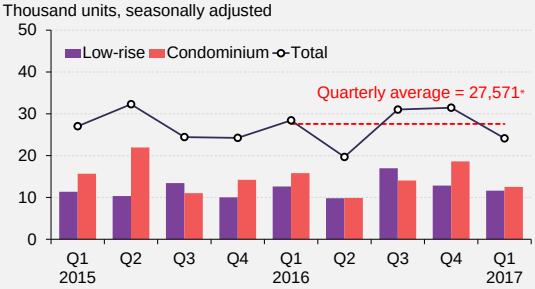
Note: *Monthly average during 2010-2016 excluding periods with government stimulus measures (November 2015 – April 2016)

Source: Bank of Thailand

Chart 2 Real estate developers deferred the launches of new projects.

New residential projects launched in Bangkok and vicinities

Thousand units, seasonally adjusted



Note: March 2017 data was proxied by previous month data.
*Monhtly average during 2010-2016 excluding periods with government stimulus measures (November 2015 – April 2016)

Source: Agency for Real Estate Affairs (AREA), calculations by the Bank of Thailand

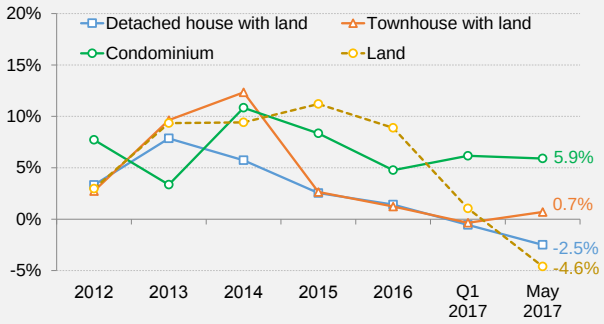
In terms of **pricing**, prices of low-rise units remained stable in the first quarter of 2017, while prices of condominium units rose albeit at a slower pace relative to those observed during 2014-2015 **(Chart 3)**. Going forward, it seemed less likely that prices of condominium units would accelerate as much as it did in the previous period for various reasons. In particular, real estate inventory remained high, and financial institutions were more cautious in extending loans to both property developers and mortgage applicants. Meanwhile, short-term speculation in condominiums through buying and reselling of property reservation remained low. A survey of property developers also revealed that the ratio of individuals trading the property reservation for speculative purposes trended down. Moreover, there were less incentives for buy-to-let property investment, as high real estate inventory was a drag that led to declines in rental yields **(Chart 4)**.

[8] Monthly average during 2010-2016 excluding the period with government real estate stimulus measures (November 2015– April 2016)

Chart 3 Condominium prices rose at a slower pace.

Growth rate of real estate price indices for Bangkok and vicinities

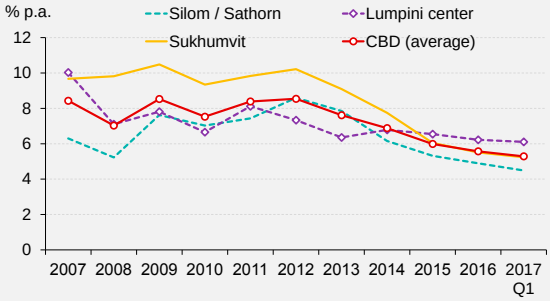
Percentage change from the same period last year



Source: Bank of Thailand

Chart 4 Rental yields on condominiums trended down.

Rental yields on condominium rentals in central business district (CBD)



Note: Condominium rentals were proxied from average rentals of grade A serviced apartment in CBD areas.

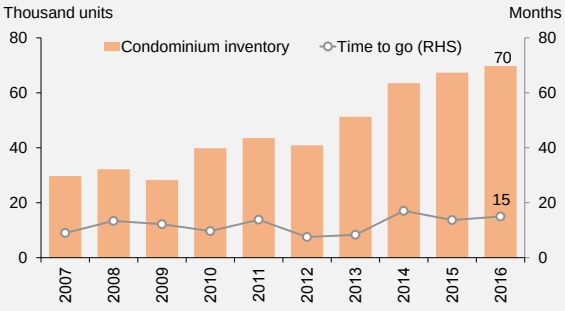
Source: CBRE and calculations by the Bank of Thailand

However, given slow demand growth, unsold condominium units[9] in Bangkok and vicinities continued to rise **(Chart 5)**. Nevertheless, the assessment of whether the current level of oversupply would pose risks to the economy must also take into account the "time to go".[10] This was because the time taken to sell all unsold units would indicate if demand for condominium was consistent with the supply in the market. A shorter time to go would reflect that the market was still in balance with sufficient demand to absorb additional supply

Latest data as of the end of 2016 suggested that there were 69,798 unsold units, which was higher than the 3-year average of 66,894 units. However, time to go stood at 15 months, which was close to the 3-year average and slightly higher than the 10-year average of 12 months. Therefore, the **oversupply of condominiums in Bangkok and vicinities was not yet a concern. Nonetheless, there remained risks for condominiums in certain areas and price ranges that warranted close monitoring**, such as 2-3 million baht units along the MRT Purple Line (Khlong Bang Phai - Bang Sue) and 1-2 million baht units along the Green Line Extension (Bearing - Samutprakarn) **(Chart 6)**. This was due to **high levels of unsold units and relatively longer time to go than in the previous year**.

Chart 5 Condominium inventory increased but 'time to go' remained close to the previous level.

Condominium inventory in Bangkok and vicinities and 'time to go'



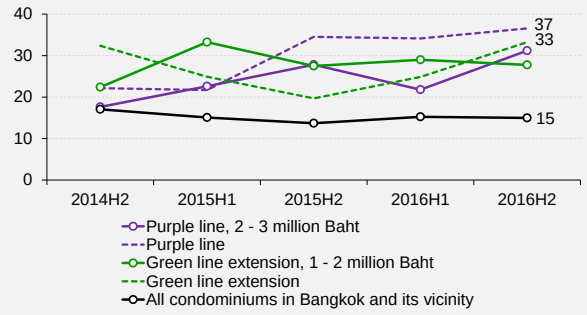
Note: 'time to go' is the time taken for all real estate inventory to be sold, using average sales volume of the past 12 months.

Source: AREA and calculations by the Bank of Thailand

Chart 6 Condominiums in certain areas and price ranges saw longer 'time to go'.

Average 'time to go'* of condominiums along the mass transit system route

Months



Note: 'time to go' is the time taken for all real estate inventory to be sold, using average sales volume of the past 12 months.

Source: AREA and calculations by the Bank of Thailand

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9 Unsold units measure the difference between total supply of condominium units (including projects of all construction status; no construction, under construction and completed construction) and total units sold (since the trading of property reservation regardless of the project construction status)

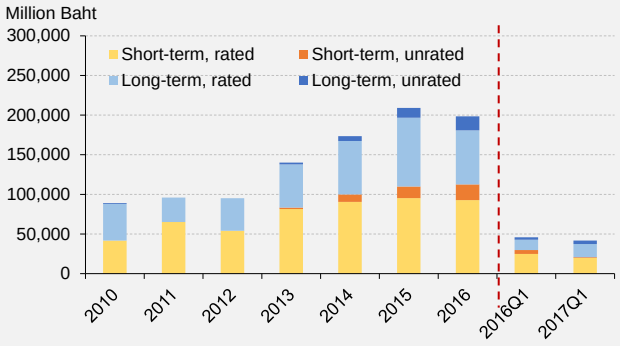
10 Time to go is calculated by the speed or the time taken to sell all unsold units in the market. This is based on assumptions that no outstanding supply is added to the market and that sales volume equals to the 12-month average. The longer time to go reflects higher risks from an oversupply in the property market.

When considering the impact of unsold units on financial positions of developers, it was found that the **increasing number of unsold units remained at manageable levels** as most unsold units belonged to large listed companies with business expertise and strong financial positions. Furthermore, some developers adjusted their business models to the changing environment. For example, some developers increased the proportion of recurring income to cushion against income shocks during the economic slowdown. Some developers also partnered with foreign companies to upgrade their business potential. Such adjustments would ensure sufficient liquidity for developers during the waiting time to sell all unsold units.

Going forward, despite the improving economic outlook, the property market would likely recover at a slower pace than expected, giving rise to the number of unsold units in the market. This was because household purchasing power was still under pressure from high levels of debt. Financial institutions were also cautious in extending loans for both property developers and mortgage applicants due to deteriorating credit quality. Nevertheless, growing demand for condominiums in Thailand by non-residents would be closely monitored as it might have implications for price speculation and acceleration in property prices in the period ahead. Moreover, some government measures such as the Royal Decree on the Managing of the Work of Aliens B.E. 2560 (2017) might put pressures on operating costs of developers in the future. Hence, it would be necessary to monitor adjustments and financial positions of real estate developers, especially SMEs with low liquidity, and debt serviceability of some developers with fragile financial positions and raised funds through issuing bills of exchange (B/E). Should developers be unable to repay debt and roll over maturing bonds, investor confidence could be dampened and thus funding costs might also rise for other businesses in the same industry **(Chart 7)**. Thus, the Committee would closely monitor the situation in the property market, developers' adjustments, and potential spillovers to the financial system going forward.

Chart 7 Real estate developers raised funds through short-term debt instruments more than in the past.

Value of new debt instrument issuances from real estate developers, classified by maturities and credit ratings



Source: ThaiBMA and calculations by the Bank of Thailand

2.2 Outlook for the Thai economy

Under the Committee's assessment, the Thai economy was projected to gain further traction and attain higher growth rates of 3.5 and 3.7 percent in 2017 and 2018, respectively, compared with the assessment in the previous *Monetary Policy Report*. The main drivers of growth were expected to be (1) improvements in exports of goods and services thanks to trading partners' economic expansion, (2) sustained fiscal impulse despite delays in some investment projects, and (3) the gradual recovery of private consumption. Meanwhile, inflation was projected to edge up overall albeit at a slower pace than previously expected due to supply-side factors.

Table 2.2 Summary of forecasts

Percent	2016*	2017	2018
GDP growth	3.2	3.5 (3.4)	3.7 (3.6)
Headline inflation	0.2	0.8 (1.2)	1.6 (1.9)
Core inflation	0.7	0.6 (0.7)	0.9 (1.0)

Note: * Outturn
() Monetary Policy Report March 2017
Sources: NESDB, Ministry of Commerce, Bank of Thailand's estimates

Summary of key forecast assumptions

- Trading partner economies were projected to record higher growth rates in 2017 than previously assessed as growth outturns in the first quarter were better than expected for most countries. This was particularly the case for the euro area where growth was supported by a continued recovery in domestic demand, and some Asian economies such as China and Malaysia that recorded stronger growth.
- The federal funds rate was expected to be raised one more time over the remainder of 2017 in addition to the two previous hikes in the first half of the year. It was expected that the Fed would also start reducing its balance sheet in accordance with the announced plan. The number of policy rate hikes in 2018 was projected to total three times.
- Asian currencies (excluding the Chinese yuan) were stronger than previously assessed given their appreciation in the second quarter of 2017. Going forward, however, they were projected to weaken in response to increases in the federal funds rate.
- The Dubai crude oil price was revised down from the previous assessment. This downward revision was due to the larger-than-expected increase in oil supply from shale oil production in the U.S., and also from Libya and Nigeria which were both exempted from the production cut agreement between OPEC and non-OPEC producers.
- Metal prices were expected to decline in the short run due to the higher-than-expected increase in iron ore inventory. Their overall outlook, however, remained close to the previous projection given that underlying fundamentals remained largely unchanged

- Public spending at current price was revised down from the previous assessment due to unexpected delays in public investments. Specifically, these included implementation delays in some of the State Railway's projects, and delayed disbursement of the Government's province funds. Meanwhile, public consumption remained largely unchanged compared with the previous assessment

- Farm income improved on account of higher outputs of second-crop rice and fruits thanks to favorable weather conditions and rainfall which returned to normal levels. The improvements seen in farm income however, was partly offset by the fall in agricultural prices that accompanied higher agricultural output.

- Public spending at current price was revised down from the previous assessment due to unexpected delays in public investments. Specifically, these included implementation delays in some of the State Railway's projects, and delayed disbursement of the Government's province funds. Meanwhile, public consumption remained largely unchanged compared with the previous assessment.

Table: Summary of forecast assumptions

Annual percentage change	2016*	2017	2018
Dubai crude oil price (U.S. dollar per barrel)	41.4	50.9 (52.3)	52.8 (54.8)
Metal prices (% YoY)	-5.3	18.6 (21.4)	0.5 (0.2)
Farm income (% YoY)	2.2	6.5 (5.4)	4.4 (4.3)
Government consumption at current price (billion baht)[1/]	2,456	2,572 (2,570)	2,688 (2,687)
Public investment at current price (billion baht)[1/]	936	1,034 (1,082)	1,163 (1,202)
Fed funds rate (% at year end)	0.63	1.38 (1.38)	2.13 (2.13)
Trading partners' GDP growth (% YoY)[2/]	3.1	3.4 (3.3)	3.3 (3.3)
Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)[3/]	154.8	157.6 (160.2)	159.8 (161.7)

Notes: [1/] Assumption includes spending on infrastructure investment plans
[2/] Weighted by each trading partner's share in Thailand total exports
[3/] Increasing index represents depreciation, decreasing index represents appreciation
* Outturns
( ) Monetary Policy Report March 2016

Merchandise exports were projected to continue expanding and more broad-based.

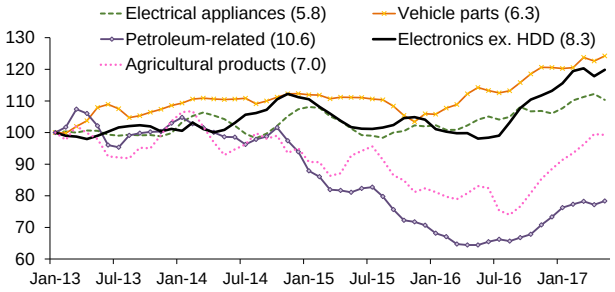
The value of merchandise exports was projected to pick up further from the pace assessed in the previous *Monetary Policy Report* and achieve 5.0 percent growth in 2017 before slowing down to 1.7 percent growth in 2018. On the whole, this represented a large improvement over the 0.1 percent growth in 2016. The ongoing global economic recovery contributed to the **more broad-based growth of Thai merchandise exports this year, both in terms of product groups and export destinations** (Chart 2.14). In terms of product groups, growth was seen in exports of electronics, electrical appliances, automobile parts, and agricultural goods such as rubber and sugar. In terms of export destinations, exports to advanced and regional economies both expanded. Notably, Thai exports of electronics benefited from the gaining popularity of smart phones and Internet of Things (IoT) products, as well as the relocation of production bases to Thailand for products such as smartphones and hard disk drives. Moreover, Thai manufacturers steadily expanded their production capacity to accommodate the growing demand for hard disk drives used in data

centers. Not only was export volume projected to increase, but export prices were also projected to rise in tandem with crude oil prices especially for commodities, for example, rubber and oil-related goods such as petroleum products. Nevertheless, the high exports value growth was still concentrated among the electronics product group which possessed high import content, and thus imports of intermediate goods increased in tandem with the exports value in this product group. As a result, the exports growth seen so far did not yield much benefit to manufacturing of goods in other product groups.

Chart 2.14 Exports of goods exhibited a stronger recovery across many product categories, especially electronics.

Value of exported goods

Seasonally adjusted index, 3-month moving average
(January 2013 = 100)



Note: Number indicated in () denotes share to total exports in 2016
Source: Customs Department, calculations by the Bank of Thailand.

In the latter half of 2017 and throughout 2018, merchandise exports would likely expand at a more gradual pace compared with the first half of 2017. This could be attributed to the high base last year, and the slowdown in exports of some products due to the slowdown in the Chinese electronics industry. In particular, China's production of smart phones was already frontloaded in earlier periods and thus its subsequent deceleration might impact Thai electronics exports which belonged to the same supply chain. In addition, rubber exports to China might decelerate as China unwound its large stock of rubber. Concurrently, there remained structural problems in certain industries such as automobiles whereby manufacturers had expanded their production capacity outward to neighboring countries. Looking ahead, key risks to the exports outlook were uncertainties pertaining to the U.S. foreign trade policies, China's potential economic slowdown as a result of ongoing economic structural reforms, as well as possibilities of lower-than-expected global economic growth.

Exports of services recovered swiftly and were projected to expand further.

Exports of services were projected to expand on the back of continued improvements in the tourism sector. The Committee revised up the projection for the number of foreign tourists to 34.9 million in 2017, and 37.3 million in 2018. This upward revision was attributed to the number of Chinese tourists which recovered swiftly from the impact of measures to curb illegal tour operators rolled out toward the end of last year, partly thanks to the launches of new direct flight routes from China to major tourist destinations in Thailand. Moreover, Thai tourism also benefited from the Chinese government's restriction on package tours to South Korea, which diverted a portion of Chinese tourists to Thailand. Meanwhile, the global economic recovery helped support increases in the foreign tourist numbers for other nationalities as well as increases in tourism spending per head. In addition,

the impact of terrorism in Europe on tourist numbers in Thailand was marginal and temporary in its effect as European tourists tended to plan their travels long ahead of time and thus trip cancellations were minimal.

Given improvements in the value of merchandise exports, growth in tourism receipts from the higher number of foreign tourists and spending per head, and the decline in the value of import due to the fall in crude oil prices, **Thailand's current account would likely record a larger surplus than previously assessed. Hence, the projection for the current account surplus was revised up to 39.7 billion U.S. dollars in 2017 and 32.7 billion U.S. dollars in 2018.**

The public sector continued to support the ongoing economic recovery.

Public spending remained an important driver of growth. Budget disbursement for both public consumption and public investment was well on track despite unexpected delays in certain investment projects of the central government such as the Pracharat internet project[11] and the provincial development funds[12], which were part of the additional budget for fiscal years 2016 and 2017. Similarly, investment figures for the Bang Yai - Kanjanaburi and Bang Pa In - Nakorn Ratchasima motorway projects were held back by the pending request for additional budget as well as the decline in construction costs from lower oil prices. Investments by state-owned enterprises were also expected to be delayed as the Red Line suburban train project faced difficulties in accessing construction sites, while some mass transit lines and the second phase of the dual-track railway project were still under the approval and contract-writing processes. Nevertheless, material progress in the Thailand-China high-speed train project had some positive contributions to the state-owned enterprise investment figures across the forecast horizon.

Private consumption was projected to expand at a gradual pace.

Private consumption was projected to expand at a gradual pace in the period ahead, supported by improvements in farm income due to higher agricultural output and prices, and expected upticks in employment in both service and export-oriented manufacturing sectors. These supporting factors were expected to also help shore up households' confidence regarding their future income somewhat. However, the outlook for private consumption remained soft as the overall income of households in the manufacturing sector did not benefit much from the exports recovery. Such limited spillovers could be attributed to the fact that employment rose in only a few industries such as food, electronics, and fabricated metals, combined with the wider use of automation in place of human labor. Meanwhile, average labor income in the manufacturing sector also remained stagnant and thus households' purchasing power did not improve markedly. In addition, high level of household debt led to a more cautious spending pattern especially among lower income households with a high debt-to-income ratio.

[11] The Pracharat internet project aimed to upgrade Thailand's telecommunication infrastructure in support of economic growth by expanding the coverage of high-speed internet via fiber optic network (FTTx). The project was part of the additional budget for 2016.

[12] The provincial development funds project was formerly known as the Pracharat project. The project was part of the additional budget for 2017.

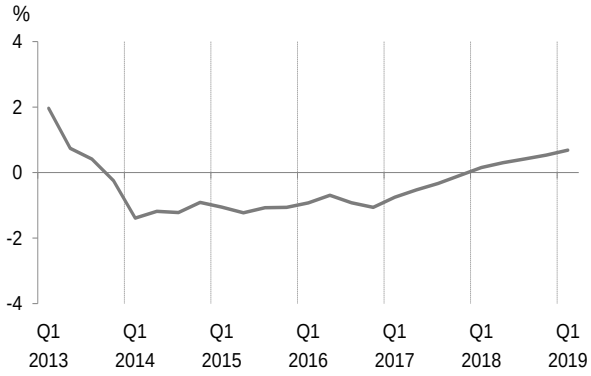
Private investment was projected to slowly recover.

Private investment showed recovery signs in some businesses. In the near term, investment expenditure to expand production capacity was observed among export-oriented manufacturing sectors, especially for electronics where there was not much production capacity left thanks to growing exports. For other sectors, however, there were no clear signs of investment expenditure partly due to remaining production capacity. Businesses were also awaiting a more robust and sustained recovery in both domestic and foreign demand, and greater clarity in the government's stimulus policies especially those regarding some infrastructure investment projects. Meanwhile, construction investment was projected to continue declining given the slowdown in real estate activities and the fact that the government's large-scale infrastructure investment projects were not yet fully implemented. However, private investment going forward would likely benefit from improvements in private consumption and exports, and from progress and greater clarity in public investment projects as well as the passing of Eastern Economic Corridor (EEC) Bill.

Inflation remained low in the near term due to supply-side factors but was projected to gradually rise going forward.

In the recent period, the Thai economy had gained traction from improvements in external demand which benefited exports of both goods and services, while the recovery in domestic demand remained soft. However, once taken into account the impact of supply-side factors especially the decline in fresh food prices that resulted from last year's high base following the drought, coupled with this year's higher agricultural output thanks to favorable weather conditions, inflation was projected to remain stable at levels below those projected in the previous *Monetary Policy Report* over the near term.

Chart 2.15 Output Gap



Looking ahead, inflation was expected to slowly rise as **demand pressures** gradually strengthened in tandem with the economic recovery as reflected by the closing output gap (Chart 2.15). Meanwhile, **cost-push inflation** would likely rise at a slower pace than expected given the decline in energy prices due to higher supply of crude oil as well as the slower increase in fresh food prices. The latter was largely attributable to (1) cyclical factors such as the low and stable food commodity prices in global markets which reduced the costs of animal feeds domestically, and (2) structural factors such as the development of new manufacturing production and irrigation technology. (BOX: *Impacts of cyclical and structural factors on inflation dynamics)*

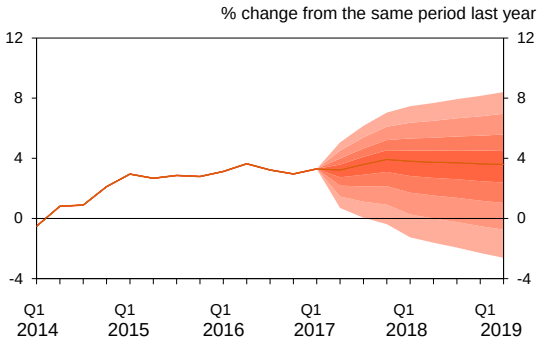
The Committee assessed headline inflation to slowly rise and gradually return to the target band in the period ahead. Nevertheless, the timing of the headline inflation's return to the target would be slightly later than previously assessed because of supply-side factors, namely the lower-than-expected outturn of fresh food prices. Meanwhile, medium and long-term inflation expectations of the public remained close to the mid-point of the target. **Given**

these developments, the Committee revised down the projections for headline inflation to 0.8 percent in 2017 and 1.6 percent in 2018. The projections for core inflation were also revised down to 0.6 percent in 2017 and 0.9 percent in 2018.

Growth projection faced more downside risks and was largely unchanged from the previous assessment. In contrast, inflation projections faced more upside risks.

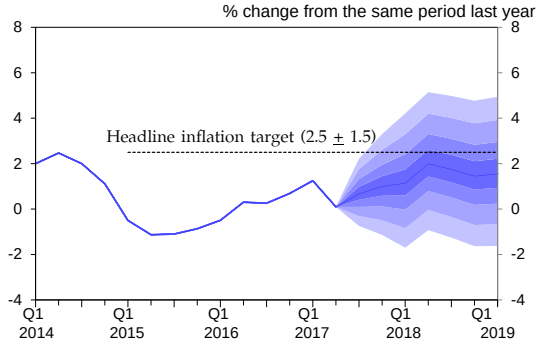
Under the Committee's assessment, the latest growth projection possessed a smaller degree of uncertainties, but the balance of risks remained tilted to the down side and was largely unchanged from the previous assessment as reflected by the growth forecast fan chart which still skewed downward throughout the forecast horizon (Chart 2.16). Despite improved outlook for trading partners' growth and global trade, the Thai economy still faced downside risks in the medium term from uncertainties surrounding the U.S. foreign trade policies that might affect the global economic outlook and lead to a significant slowdown in Thailand's trading partner economies. Other downside risks included China's ongoing economic structural reforms and geopolitical risks, both of which could impact trading partners' growth and the recovery of domestic purchasing power which was not yet sufficiently broad-based. On the upside, there were possibilities that growth outturns might be higher than the baseline projection given that the U.S. economy might achieve stronger growth from its domestic stimulus which could in turn translate to better-than-expected growth for Thailand's trading partner economies. In contrast to the balance of risks to growth, **the balance of risks to inflation tilted to the upside (Charts 2.17 and 2.18)** as more stringent regulations on immigrant workers could further tighten labor market conditions and in turn raise labor costs for businesses. The extent in which labor costs would rise, however, would depend on the implementation and management by the authorities. Another upside risk to inflation was the excise tax increase to be implemented for products such as beer and sugary drinks.

Chart 2.16 Growth forecasts



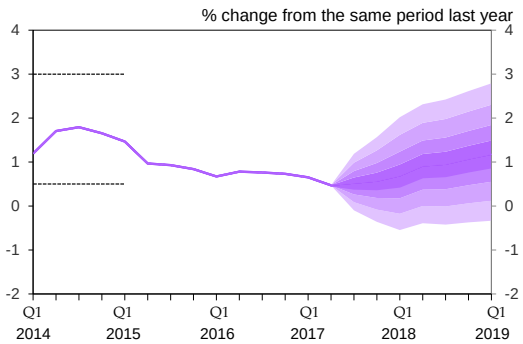
Note: Fan chart covers 90% of the probability distribution

Chart 2.17 Headline inflation forecasts



Note: Fan chart covers 90% of the probability distribution

Chart 2.18 Core inflation forecasts



Note: Fan chart covers 90% of the probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2016*	2017	2018
GDP growth	3.2	3.5 (3.4)	3.7 (3.6)
Domestic demand	2.7	3.0 (3.2)	3.4 (3.3)
Private consumption	3.1	3.1 (2.7)	3.1 (3.1)
Private investment	0.4	1.7 (2.4)	3.6 (3.7)
Government consumption	1.7	2.2 (2.2)	1.9 (1.9)
Public investment	9.9	7.7 (11.8)	9.2 (7.5)
Exports of goods and services	2.1	4.6 (2.1)	2.7 (2.8)
imports of goods and services	-1.4	4.8 (2.6)	2.6 (2.9)
Current account (billion, U.S. dollars)	47.7	39.7 (36.9)	32.7 (33.1)
Value of merchandise exports	0.1	5.0 (2.2)	1.7 (2.0)
Value of merchandise imports	-5.1	10.9 (7.2)	5.4 (5.3)

Note: *Outturns

( ) Monetary Policy Report March 2016

Impacts of cyclical and structural factors on inflation dynamics

Headline inflation was expected to trend up on account of cyclical factors stemming from the steady recovery in the global and Thai economy. However, the increase was expected to be gradual due to downward pressures from structural factors.

The recent decline in Thailand's headline inflation was due mainly to supply-side factors particularly oil prices. This was reflected by the fact that the average headline inflation over 2000 – 2014 was higher than the average level during 2015 – 2016, where a structural shift in crude oil prices production was observed **(Chart 1)**. **However, headline inflation over the first six months of 2017 continued to improve from the previous year despite remaining at the low level of 0.67%**. This improved inflation trend was attributable to accelerating energy inflation from the previous year as the global economic recovery underpinned rising crude oil prices. Moreover, an increase in fuel adjustment charge (ft)[13] and the low base effect from last year's oil prices also contributed to higher inflation. Nonetheless, **this year's low headline inflation was mainly attributed to fresh food inflation, especially vegetables and fruits, that fell** on account of higher agricultural output thanks to favorable conditions coupled with the high base following last year's drought. On the other hand, **core inflation (excluding fresh food and energy) slowed down** as the base effect of last year's increase in the excise tax on tobacco waned. However, excluding the effect of the change in the tobacco tax, core inflation still remained at a low level given the low pass-through of production costs to prices and low demand-pull inflationary pressures.

Chart 1 Headline inflation softened due to lower fresh food prices

Contributions to headline inflation (Percent)	Historical average (2000 – 2014)1/	Historical average (2015 – 2016)	2015	2016	2017 2/
Fresh Food	0.80	0.24	0.11	0.41	-0.23
Energy	0.74	-1.20	-1.76	-0.77	0.49
Core inflation	1.04	0.61	0.75	0.55	0.41
Headline inflation	2.58	-0.35	-0.90	0.19	0.67

Note: 1/ Calculated over the period during which Thailand adopted inflation targeting as its monetary policy framework until the period before structural changes in global crude oil prices
2/ Averaged in the first six months of 2017

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

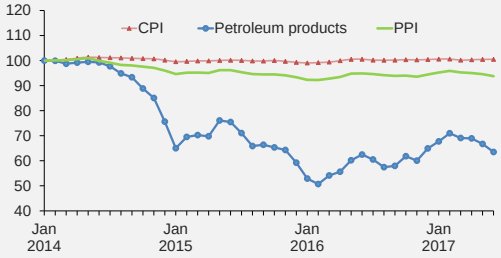
[13] The Energy Regulatory Commission (ERC) increased the Ft charge in May – August by 12.52 satang/unit due to rising costs of natural gas used to generate electricity.

Thailand's inflation developments was consistent with the global trend. Headline inflation in Thailand and the global economy trended up from last year in line with energy prices. However, food inflation declined in Thailand as well as in some other countries such as China, Hong Kong, and Taiwan. The main reasons behind the declining were that prices of animal feeds were stabilized at a low level in line with the prices in the global market and there were no natural disasters as in the previous year. Meanwhile, global and Thailand's core inflation stayed at low levels, reflecting domestic demand that had yet to fully recover.

Looking ahead, although headline and core inflation projections were revised down from the previous Monetary Policy Report, inflation was projected to gradually rise in the second half of 2017 and throughout 2018 following the upward trends in cyclical factors. The downward revision this year was based on outturns in the first half that turned out lower than expected on the back of declining fresh food prices especially vegetables and fruits and the downward revision of oil price projections due to the higher-than-expected global oil stock.[14] On the other hand, **inflation was projected to rise in the second half of 2017 and over 2018 owing to upward trends in cyclical prices**. In particular, crude oil prices were projected to rise along with global economic recovery. Also, the fuel adjustment charge (ft) was expected to increase due to the rise in costs of natural gas used to generate electricity. As a result, energy inflation would consequently edge up. Meanwhile, core inflation was expected to slightly increase following the prices of the prepared food component in core inflation that were expected to gradually increase along with input costs owing to a number of factors. These included domestic oil prices that expected to rise following the global market and fresh food prices that were expected to increase as the base effects would eventually dissipate. Non-food-in-core inflation was projected to slowly rise as the domestic demand continually recovered. Nonetheless, short-term core inflation was projected to gradually increase. As positive spillovers from export recovery to domestic demand were limited, demand-pull inflationary pressures still remained relatively low. At the same time, the cost-price pass-through, especially from higher energy prices, was also limited as producers could absorb higher production costs, without adjusting prices. This was partly because producers benefited from having not adjusted prices down during the earlier fall in production costs **(Chart 2)**. In addition, despite the closing of the output gap in early 2018, inflation persistence was expected to lead to a gradual increase in core inflation.

Chart 2 The pass-through effects of higher production costs remain low

Consumer Price Index (CPI) Producer Price Index (PPI) and Petroleum products Index (January 2014 = 100)



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Despite an upward trend in cyclical prices following the recovery in the global and Thai economy, structural factors still weighed on inflation and partly caused the pace of inflation increase to be gradual. Key structural factors were as follows. **First, technology for shale oil production** was being developed to reduce the costs of oil production. This led the stock of crude oil to be more responsive to demand, resulting in low global price as well as retail domestic oil prices. **Second, new methods of using larger agricultural areas** could lower costs of production because of economies of scale. In addition, **developments in production technology and irrigation system** could slow down the rises in production costs as well as fresh food prices. **Third, globalization** induced price competition in both good and services. Moreover, global value chain

[14] The high level of shale oil stock and rising production levels in Libya and Nigeria after resolving domestic unrest led to a higher level of global crude oil stock, despite an extension of the production cut agreement by OPEC and non-OPEC producers by another nine months.

expansion allowed many countries to easily access to raw materials at cheaper prices. As domestic firms had less room to increase their prices of goods, core inflation was being kept low. **Fourth, Thailand's population structure changed toward an aging society**. The United Nations (UN) projected that the proportion of Thai people who were more than 60 years old would increase to 20 percent in 2021. Despite inconclusive results on the study of impact of aging on inflation, there were several channels through which population aging could affect inflation and each of the channels can affect in the different directions.[15] In the case of Thailand, population aging could weigh down on inflation as income of the elderly might not be sufficient for retirement. In fact, savings for retirement of working-age population were still limited. Consequently, this could led private consumption to expand in a gradual pace and the increase in inflation to be gradual. **Fifth, E-commerce was on the rise**. An international study found that such factor could affect inflation via two channels: lower production costs, and price transparency which could limit the ability of businesses to raise prices. For Thailand, the direct impact of E-commerce was difficult to measure given the small size of E-commerce items in the inflation basket. Indirect impact from price transparency, however, was relevant as it was more difficult for businesses to increase prices. **The Committee would continue to closely monitor developments and study the impact of structural factors on inflation going forward**.

[15] Population aging was expected to lower economic growth and inflation. Investment could decline because of a smaller domestic labor force and a higher saving rate for retirement (IMF, 2014). On the other hand, population aging could put upward pressures on inflation because of excess demand, as consumption was expected to be greater than production capacity (Juselius and Takats, 2015).

3. Monetary Policy Decision

Thailand's economic growth continued to gain further traction but the positive effects were not yet broad-based. Therefore, an accommodative monetary policy stance was still warranted.

The Committee noted that while Thailand's growth outlook continued to improve driven primarily by external factors, the expansion was not yet broad-based. These improvements were particularly seen in higher-than-expected growth in merchandise exports and the continued recovery in the tourism sector. However, domestic demand softened as private consumption was expected to gradually expand. While farm income improved, private consumption was still weighed down by high debt levels since the drought last year. Also, the positive spillovers from export expansion to employment and household income were so far limited. Private investment was concentrated largely among export-related businesses and expected to slowly gain momentum as demand picked up, which would help to gradually reduce excess capacity. Meanwhile, public expenditure continued to be a key economic driver despite some delay in public investment projects due partly to delays in budget disbursement and difficulties in acquiring additional budget appropriations in some state-owned enterprises.

The Committee viewed that Thailand's economic outlook had improved but the benefits from the export recovery was limited to export-related industries and did not have much positive spillovers to other parts of the economy. Private investment slowly recovered. Export growth still faced significant risks from the external front, including uncertainties surrounding the U.S. economic and foreign trade policies and responses of U.S. trading partners. In addition, intensified geopolitical risks could undermine sentiment of foreign tourists and eventually Thailand's tourism sector. Thus, the Committee saw the need for continued accommodative monetary policy to support economic growth. The Committee would closely monitor the continuation of Thailand's economic growth and its distribution across sectors in order to formulate appropriate monetary policy in the period ahead. Moreover, the impact of more stringent regulations on immigrant workers on the economy, employment, and inflation would warrant close monitoring.

Headline inflation remained on an upward path, despite some slowdown from supply-side factors.

The Committee projected that the average annual headline inflation would be lower than previously assessed primarily due to supply-side factors, namely larger-than-expected falls in crude oil and fresh food prices. Hence, further monetary policy accommodation in this period would yield a limited impact on inflation. While headline inflation was expected to rise in tandem with economic growth, the increase could be limited by structural factors such as population aging and technological trends in businesses. Medium-term inflation expectations softened somewhat, but remained close to the mid-point of the target, reflecting the public's confidence on the inflation target. Moreover, the Committee noted demand-pull inflationary pressures that remained subdued was consistent with core inflation that was stable at the low level. This was because economic growth was not yet sufficiently broad-based and there remained economic slack as seen in the negative output gap and an increase in the

unemployment rate. Consequently, the recent improvement in growth did not put significant upward pressures on inflation despite increases in wages and production costs. The Committee therefore deemed accommodative monetary policy important to support domestic demand growth that would later increase inflationary pressures in the period ahead.

Financial conditions remained accommodative.

The Committee assessed that overall financial conditions were accommodative and conducive to economic growth, as reflected by high liquidity in the financial system and low real policy rate. Financing costs declined slightly after a number of commercial banks cut their benchmark retail interest rates. Moreover, lower bond yields enabled overall financial conditions in the second quarter of 2017 to be more accommodative, relative to the end of the previous quarter.

With regard to the exchange rates, the baht against the U.S. dollar was volatile and strengthened relative to the end of the previous quarter. This was in line with movements of regional currencies as investors revised their expectations regarding the Fed's policy rate hikes to be more gradual after the release of weaker-than-expected U.S. economic figures. As a result, there were increased capital flows into emerging markets including Thailand. The Committee would therefore closely monitor short-term capital flows and their impact on the exchange rates going forward.

Financial stability remained sound. However, there were pockets of risks warranted close monitoring.

The Committee viewed that overall financial stability remained sound and able to provide cushion against volatilities in global and domestic financial markets. Nevertheless, there remained pockets of risks that warranted monitoring, particularly debt serviceability of SMEs that did not yet improve, reflecting their competitiveness issues under changing economic contexts. At the same time, search-for-yield behavior in the prolonged low interest rate environment remained present as there were increased investments in riskier assets relative to bank deposits, reflecting signs of underpricing of risks. For instance, foreign investment funds (FIFs) continued to expand, but the risks they pose to the financial system remained limited as their investments were mostly in the form of deposits or short-term bonds in countries with investment-grade credit rating. In addition, investment in unrated bonds slowed down owing to earlier defaults by some issuers. Overall, the bond market underwent orderly adjustments.

Despite Thailand's sound financial stability, the Committee viewed that were still other pockets of risks that remained undetected or have interlinkages that could impact various parts of the financial system. Therefore, close collaboration among regulatory agencies would be necessary to reduce regulatory gaps and allow risks to be addressed in a more efficient and timely manner.

The Committee voted unanimously to maintain the policy rate at 1.50 percent so that monetary policy remain accommodative and conducive to the continuation of economic growth, which would in part facilitate the return of headline inflation to the target.

The Committee voted unanimously to maintain the policy interest rate at 1.50 percent at the meetings on 24 May and 5 July 2017. While Thailand's economic growth became stronger especially from external factors, domestic demand expansion was not yet sufficiently broad-based and the economy was still faced with risks on external fronts that could hamper the export and overall economy recovery. Moreover, inflationary pressures remained low as headline inflation this year might average below the target despite retaining an upward trajectory. Meanwhile, financial conditions eased and financial stability remained sound. In this light, the Committee concurred that the degree of monetary policy accommodation should be remained for some time so that economic growth could continue and become more broad-based, which would enable headline inflation to return to target in a sustainable manner. The Committee would stand ready to utilize an appropriate mix of available policy tools in order to achieve these policy goals while also ensuring financial stability.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2015	2016	2016				2017
			Q1	Q2	Q3	Q4	Q1
GDP growth	2.9	3.2	3.1	3.6	3.2	3.0	3.3
Production							
Agriculture	-5.7	0.6	-2.0	-0.4	0.9	3.0	7.7
Non-agriculture	3.9	3.5	3.7	3.9	3.2	3.2	2.9
Manufacturing	1.5	1.4	-0.2	2.2	1.6	2.2	1.2
Construction	17.0	8.3	11.9	9.9	5.2	6.1	2.8
Wholesales and retail trade	3.9	5.0	4.6	4.7	5.2	5.6	5.9
Hotels and restaurants	14.6	10.3	12.3	10.8	13.5	4.9	5.3
Transport, storage, and communication	5.1	5.5	6.4	4.2	6.5	5.2	5.9
Financial intermediation	8.8	6.2	5.9	6.2	5.8	6.7	4.4
Real estate, renting, and business activities	1.9	1.8	2.2	2.1	1.1	1.9	3.3
Expenditure							
Domestic demand	2.9	2.7	4.4	3.5	0.9	2.2	2.3
Private consumption	2.2	3.1	2.8	4.0	3.0	2.5	3.2
Private investment	-2.2	0.4	2.4	0.3	-0.8	-0.4	-1.1
Government consumption	3.0	1.6	9.1	2.4	-5.2	1.8	0.2
Public investment	29.3	9.9	12.9	12.8	5.8	8.6	9.7
Imports of goods and services	0.0	-1.4	-5.2	-2.5	-1.1	3.4	6.0
imports of goods	0.2	-2.1	-7.6	-2.8	-1.5	3.6	7.3
imports of services	-1.0	1.7	5.7	-1.3	0.5	2.0	0.8
Exports of goods and services	0.7	2.1	4.8	1.0	1.4	1.1	2.7
exports of goods	-3.4	0.0	0.8	-1.9	-0.4	1.4	2.6
exports of services	17.1	9.3	18.1	11.6	7.7	0.4	3.2
Trade balance (billion, U.S. dollars)	26.8	36.5	11.7	8.5	9.2	7.1	8.8
Current account (billion, U.S. dollars)	32.1	47.7	17.8	8.6	11.2	10.2	15.2
Financial account (billion, U.S. dollars)	-17.1	-26.5	-0.3	-3.1	-10.5	-12.7	-6.8
International reserves (billion, U.S. dollars)	156.5	171.9	175.1	178.7	180.5	171.9	180.9
Unemployment rate (%)	1.0	1.0	1.0	1.1	0.9	1.0	1.2
Unemployment rate, seasonally-adjusted (%)	n.a.	n.a.	0.9	1.0	1.0	1.1	1.1

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

Financial Stability Dashboard

Indicators	2015	2016	2016			2017		
			Q2	Q3	Q4	Q1	Apr	May
1. Financial market sector								
Bond market								
Bond spread (10 years - 2 years)	1.1	0.6	0.5	0.6	0.8	1.1	1.1	1.2
Equity market								
SET index (end of period)	1,288.0	1,542.9	1,445.0	1,483.2	1,542.9	1,575.1	1,566.3	1,561.7
Actual volatility of SET index[1/]	13.9	15.2	11.9	13.6	15.2	7.0	5.1	5.7
Price to Earnings ratio (P/E ratio) (times)	22.6	18.6	22.0	21.3	18.6	17.4	17.3	16.2
Exchange rates								
Actual volatility of Thai baht (%annualized)[2/]	5.1	4.4	4.6	3.8	5	3.53	4.13	4.09
Nominal Effective Exchange Rate (NEER)	108.5	106.2	105.1	105.8	107.0	108.7	109.6	109.5
Real Effective Exchange Rate (REER)	104.4	100.5	100.0	100.3	101.1	102.0	102.5	n.a.
2. Financial institution sector[3/]								
Minimum Lending Rate (MLR)[4/]	6.50	6.26	6.26	6.26	6.26	6.26	6.26	6.20
12-month fixed deposit rate[4/]	1.40	1.38	1.38	1.38	1.38	1.38	1.38	1.38
Capital adequacy								
Capital funds / Risk-weighted asset (%)	17.4	18.0	17.5	18.5	18.0	17.8	18.0	n.a.
Earning and profitability								
Net profit (billion, Thai baht)	192.3	199.2	50.9	49.8	47.4	51.1	n.a.	n.a.
Return on assets (ROA) (times)	0.9	1.1	1.2	1.2	1.2	1.2	n.a.	n.a.
Liquidity								
Loan to Deposit and B/E (%)	97.0	96.3	96.4	96.9	96.3	95.7	95.3	95.6
3. Household sector								
Household debt to GDP (%)	81.2	n.a	80.3	80.0	79.8	78.6	n.a.	n.a.
Financial assets to debt (times)	2.6	n.a	2.6	2.6	2.62	n.a.	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)								
Consumer loans	2.6	2.7	2.6	2.7	2.7	2.8	n.a.	n.a.
Housing loans	2.4	2.9	2.7	2.8	2.9	3.2	n.a.	n.a.
Auto leasing	2.3	1.8	1.9	1.8	1.8	1.6	n.a.	n.a.
Credit cards	4.0	3.7	4.2	5.1	3.7	3.8	n.a.	n.a.
Other personal loans	2.7	2.9	2.7	2.9	2.9	2.9	n.a.	n.a.
4. Non-financial corporate sector[5/]								
Operating profit margin (OPM) (%)	7.4	8.3	8.7	8.2	7.8	8.4	n.a.	n.a.
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.7	0.7	n.a.	n.a.
Interest coverage ratio (ICR) (times)	5.7	6.6	7.1	5.9	6.8	6.2	n.a.	n.a.
Current ratio (times)	1.7	1.6	1.7	1.7	1.6	1.6	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)								
Large businesses	1.6	1.5	1.9	1.9	1.4	1.5	n.a.	n.a.
SMEs	3.5	4.3	3.8	4.0	4.3	4.5	n.a.	n.a.

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2015	2016	2016			2017		
			Q2	Q3	Q4	Q1	Apr	May
5. Real estate sector								
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)								
Total	59,667	61,452	15,611	14,149	16,205	12,244	3,166	4,974
Single-detached and semi-detached houses	13,152	13,409	3,503	3,216	3,179	2,802	758	1,176
Townhouses and commercial buildings	19,210	20,187	5,410	4,718	4,967	4,315	1,149	1,695
Condominiums	27,305	27,856	6,698	6,215	8,059	5,127	1,259	2,103
Number of new housing units launched for sale (Bangkok and Vicinity) (units)								
Total	107,988	110,575	19,680	31,017	31,452	24,138	5,440	7,063
Single-detached and semi-detached houses	17,637	19,433	3,099	6,328	4,973	1,916	349	963
Townhouses and commercial buildings	27,518	32,792	6,702	10,638	7,861	9,698	2,819	1,390
Condominiums	62,833	58,350	9,879	14,051	18,618	12,524	2,272	4,710
Housing price index (2009 = 100)								
Single-detached houses (including land)	129.0	130.8	133.7	131.3	128.8	128.6	128.7	130.2
Townhouses (including land)	135.9	137.6	139.1	136.8	135.6	138.3	139.1	140.2
Condominiums	158.7	166.2	162.1	169.2	173.6	169.8	170.7	169.6
Land	157.2	171.2	173.7	170.2	171.3	171.3	169.3	167.1
6. Fiscal sector								
Public debt to GDP (%)	44.4	42.2	42.8	42.8	42.2	42.2	42.6	n.a.
7. External sector								
Current account balance to GDP (%)[6/]	8.1	11.7	8.6	10.9	9.7	14.1	n.a.	n.a.
External debt to GDP (%)[7/]	32.1	32.5	35.4	34.5	32.5	33.1	n.a.	n.a.
External debt (billion, U.S. dollars)	131.4	131.3	143.1	139.4	131.3	134.8	134.9	137.4
Short-term (%)	40.0	40.2	42.5	40.2	40.2	39.5	39.2	38.9
Long-term (%)	60.0	59.8	57.5	59.8	59.8	60.5	60.8	61.1
International reserves / Short-term external debt (times)	3.0	3.3	2.9	3.2	3.3	3.4	3.5	3.4

Note:

[6/] Current account / Nominal GDP at the same quarter

[7/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2017			2018				2019
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
10-12	0	0	0	0	0	1	1	1
8-10	0	0	2	3	3	4	4	5
6-8	1	6	11	11	12	12	12	12
4-6	21	28	29	25	24	23	21	21
2-4	53	39	31	28	26	25	24	23
0-2	23	21	19	20	19	19	19	18
(-2)-0	2	6	7	10	10	11	11	12
< (-2)	0	1	2	4	5	6	7	8

Table: Probability distribution of core inflation forecast

Percent	2017			2018				2019
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
3.5-4.0	0	0	0	0	0	0	0	1
3.0-3.5	0	0	0	0	1	1	2	2
2.5-3.0	0	0	0	1	3	3	5	6
2.0-2.5	0	0	1	4	7	8	10	12
1.5-2.0	0	1	6	11	15	15	17	18
1.0-1.5	0	12	18	20	22	22	21	21
0.5-1.0	0	42	32	25	23	22	20	19
0.0-0.5	100	38	28	21	17	16	14	13
(-1)-0.0	0	8	12	12	8	8	7	6
(-2)-(-1)	0	0	3	4	3	3	3	2
< -2	0	0	0	1	1	1	1	1

Table: Probability distribution of headline inflation forecast

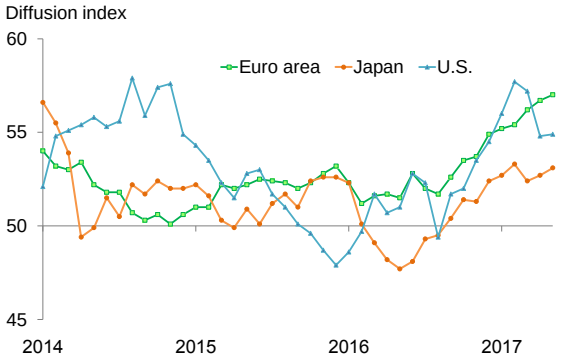
Percent	2017			2018				2019
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 7	0	0	0	0	0	0	0	1
6-7	0	0	0	0	1	1	1	1
5-6	0	0	0	2	4	4	3	3
4-5	0	0	1	5	9	8	7	7
3-4	0	1	6	10	16	14	12	13
2-3	0	8	17	17	21	19	18	18
1-2	0	30	27	22	20	21	20	20
0-1	100	41	27	20	15	16	18	17
(-1)-(0)	0	18	15	14	8	10	12	11
(-2)-(-1)	0	2	5	7	3	4	6	6
(-3)-(-2)	0	0	1	3	1	2	2	2
(-4)-(-3)	0	0	0	1	0	0	1	1
< (-4)	0	0	0	0	0	0	0	0
	100	100	100	100	100	100	100	

4.2 Chart Pack

The Global Economy

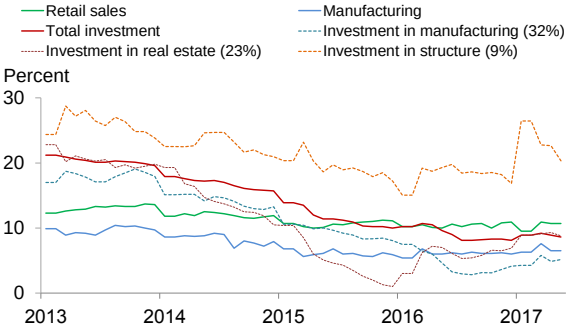
Advanced economies continued to recover on the back of improvements in domestic demand and exports following global trade recovery. China expanded well in the first quarter of 2017, but was expected to slow down in the period ahead due to the Chinese government's surveillance measures. Meanwhile, Asian exports steadily improved especially for electronics and commodities.

Manufacturing Purchasing Manager Index



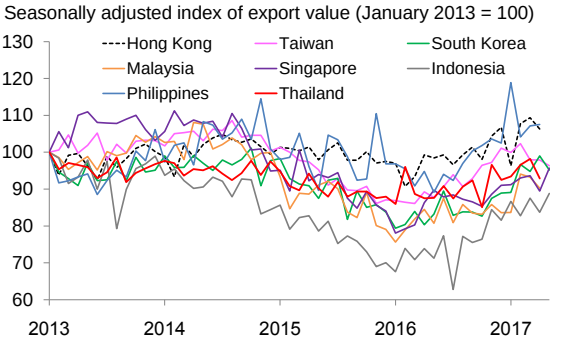
Source: Bloomberg

China's economic indicators (Change from the same period last year)



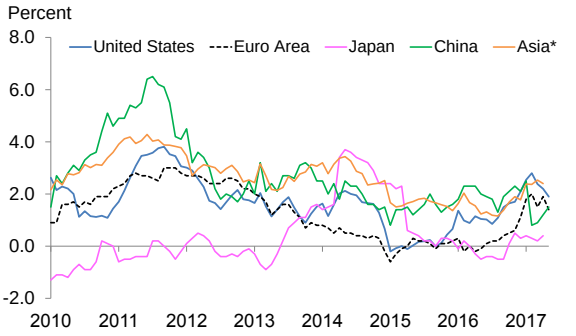
Note: ( ) denotes share to total investment
Source: CEIC

Asian exports



Source: CEIC

Inflation of Thailand's major trading partners

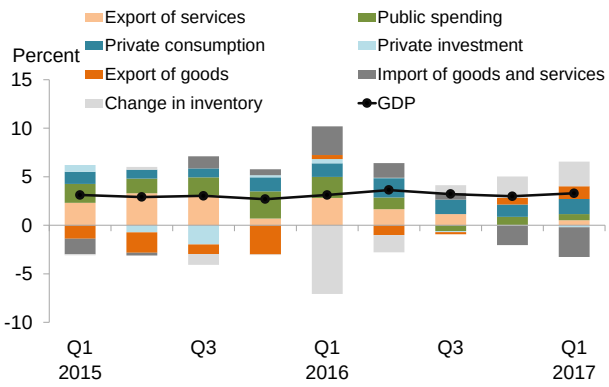


Note: * Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore and Taiwan
Source: CEIC

The Thai Economy

Thailand's economic growth gained traction in the first quarter of 2017 on account of an expansion in private consumption, particularly spending on durable goods, and growth in merchandise exports that was more broad-based. The number of foreign tourists swiftly recovered and was expected to continue expanding. Public spending slowed down after the earlier acceleration of investment expenditure.

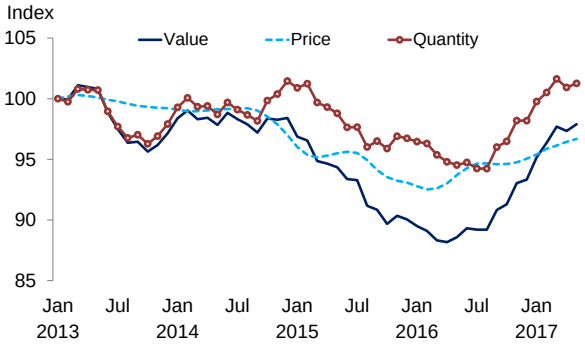
Contribution to Thailand's GDP growth[1/]



Note: [1/] Calculated by Chain Volume Measure method (CVM)

Thai exports (excluding gold): value, price and quantity

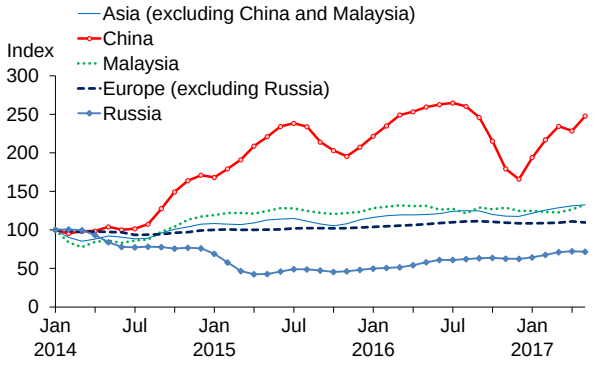
(3-month moving average, seasonally adjusted; January 2013 = 100)



Source: Customs Department and Ministry of Commerce, calculations by Bank of Thailand

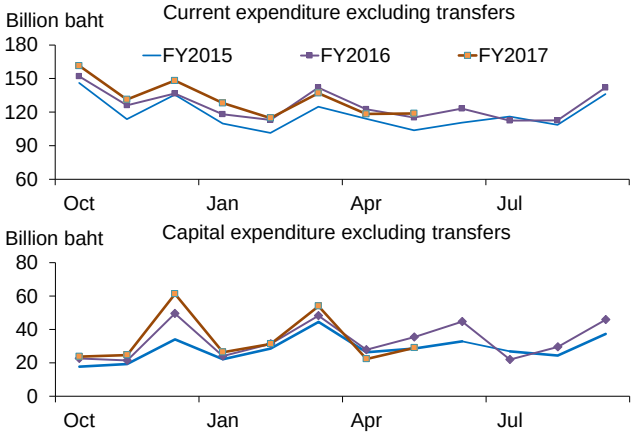
Index of foreign tourists classified by nationality

(3-month moving average, sessionally adjusted; January 2014 = 100)



Source: Department of Tourism

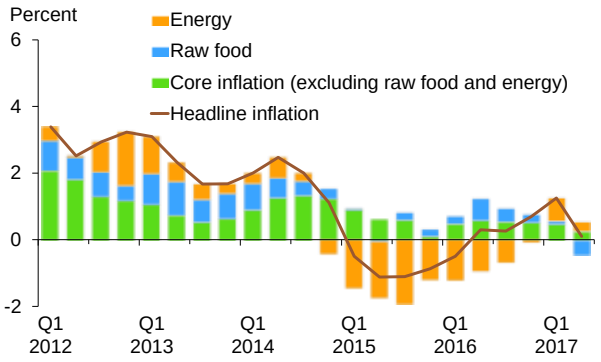
Public spending by central government



Inflation

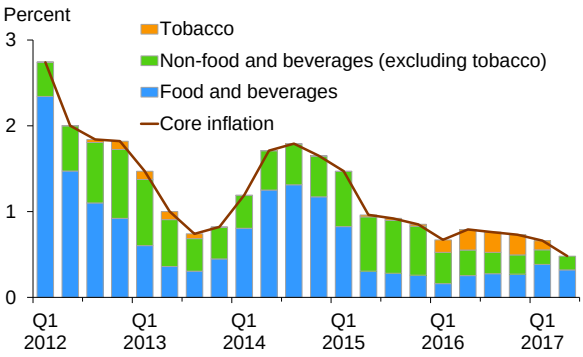
Headline inflation declined due mainly to fresh food prices following last year's high base during the drought and this year's abundant supply amid favorable weather. Core inflation slowed down owing to (1) prices of prepared food, which fell in line with those of fresh food, and (2) last year's high base due to the tobacco excise tax. Meanwhile, five-year-ahead inflation expectations by professional forecasters remained close to the midpoint of the inflation target.

Contribution to headline inflation



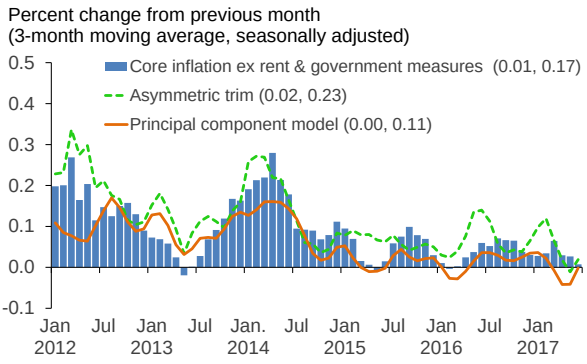
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

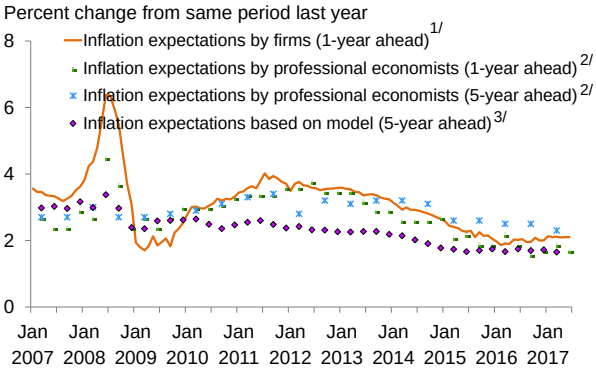
Underlying inflation indicators



Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of May 2017, while the second value is 2004 - 2014 average; Asymmetric trim excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; Principal component model calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Inflation expectations

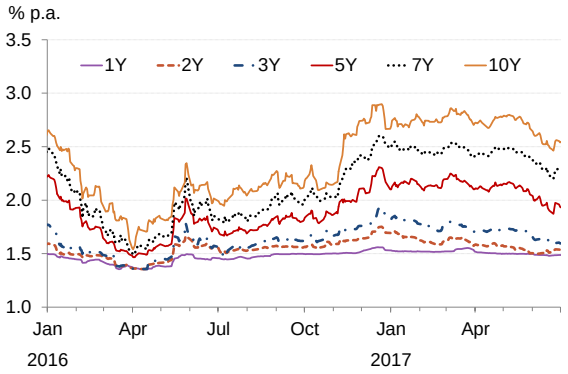


Source: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

Financial conditions

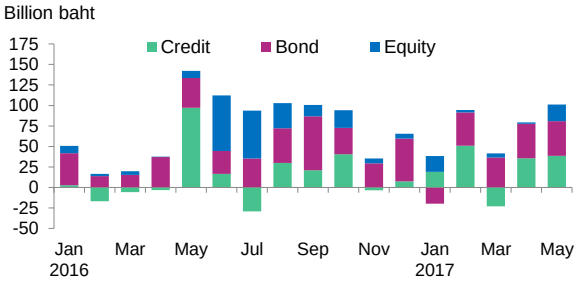
Medium- to long-term government bond yields declined due to external factors, while financing costs through commercial banks decreased following cuts in benchmark retail interest rates. Total corporate financing edged up through both credit and bond markets. The Thai baht appreciated against the U.S. dollar, consistent with movements of regional currencies, as the dollar weakened.

Thai government bond yields



Source: Bank of Thailand and Thai Bond Market Association (Thai BMA

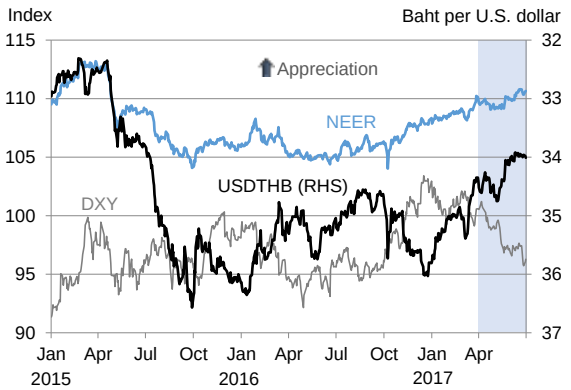
Total corporate financing by instruments*



Note: * Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bond excluding commercial banks, and newly issued equities.

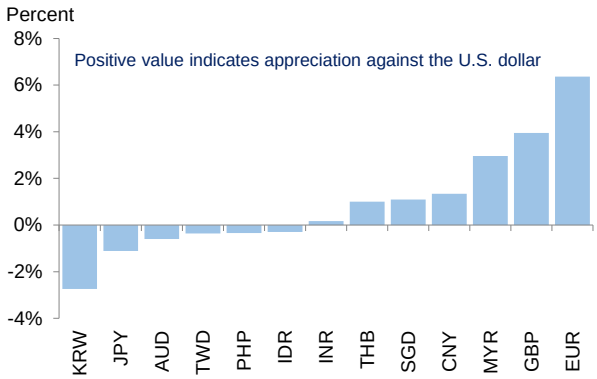
Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

The Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), and the Dollar Index (DXY)



Source: Bank of Thailand and Reuters (data as of 4 July 2017)

Currency movements vis-a-vis the U.S. dollar (4 July 2017 compared to 31 Mar 17)

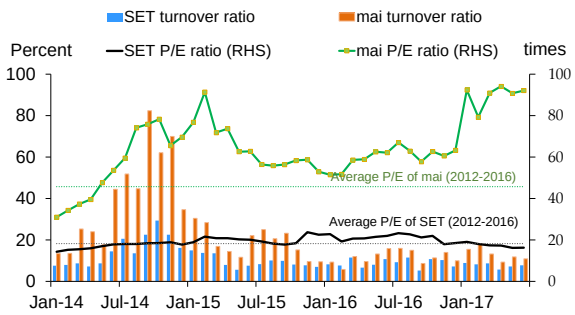


Source: Bank of Thailand and Reuters (data as of 4 July 2017)

Stability: financial markets

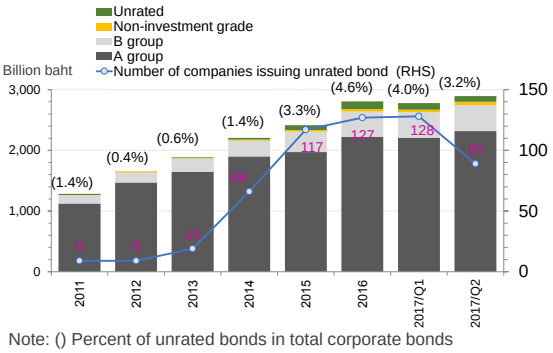
The price-to-equity ratio (P/E) of the Stock Exchange of Thailand (SET) stayed close to the historical average. The P/E ratio of the Market for Alternative Investment (mai) stabilized at the high level. The issuance of unrated bonds slowed down since the end of 2016 after some defaults but constituted a small portion in the total corporate bonds outstanding.

Current price-to-earning ratio and turnover ratio of SET and mai



Source: Stock Exchange of Thailand (as of 30 June, 2017)

Corporate bonds outstanding

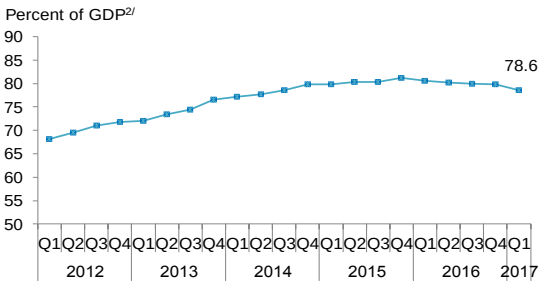


Note: () Percent of unrated bonds in total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Stability: the household sector

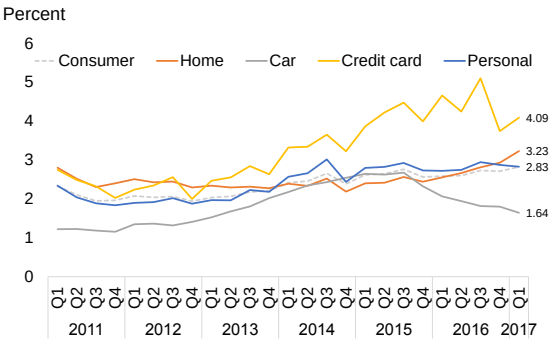
The ratio of household debt to GDP continued to trend down. However, deterioration in debt payment ability warranted close monitoring.

Household debt[1/]



Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

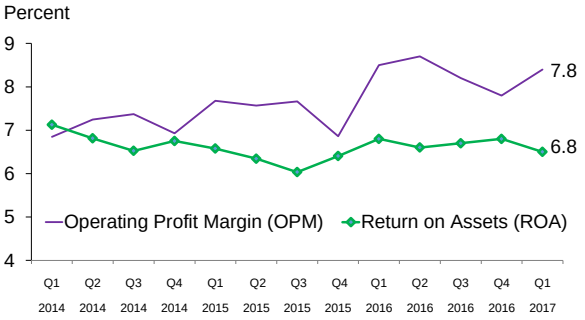


Source: Bank of Thailand

Stability: the corporate sector

Overall stability of the corporate sector remained sound. However, some businesses had yet to benefit from the economic recovery, especially small businesses as reflected by the loan quality of SMEs.

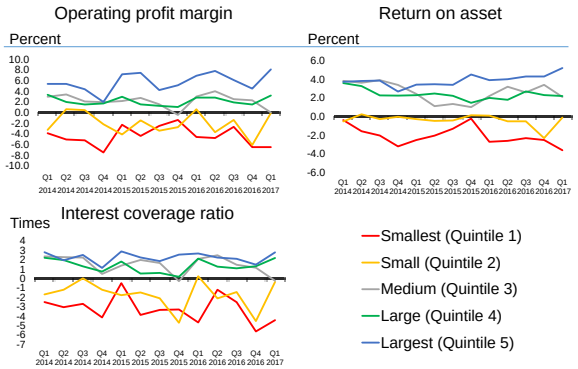
Operating Profit Margin (OPM) and Return on Assets (ROA)*



Note: * Median estimates; ROA is returns to average assets.
OPM is operating profits to total sales.

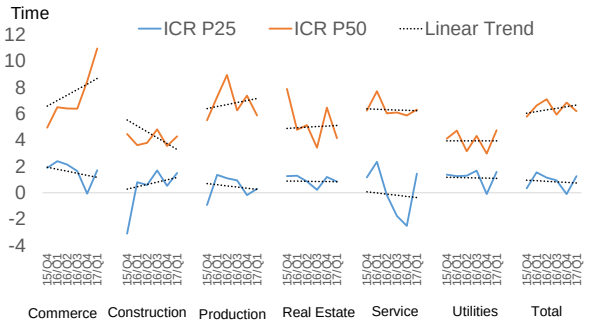
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Profitability and debt serviceability at 25th percentile of each group of firm size



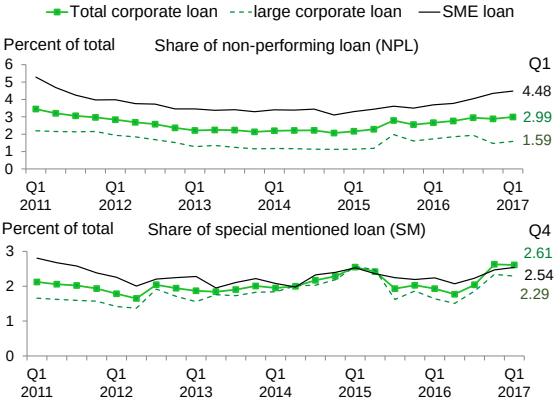
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Interest Coverage Ratio (ICR), classified by sectors



Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Loan quality of corporate sector

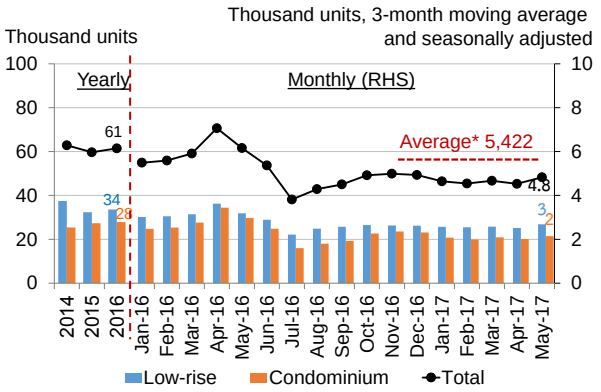


Source: Bank of Thailand

Stability: the real estate sector

Demand for residences in Bangkok and its vicinity, as indicated by new mortgage loans provided by commercial banks, remained at a low level relative to historical averages. On the supply side, developers launched more new residential projects but the time-to-go rate still remained around the same level, reflecting a largely balanced market. Residential prices rose at a slower pace.

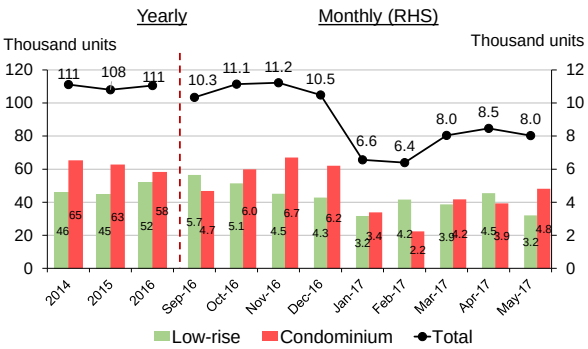
Residential units in Bangkok and its vicinity with approved mortgages by commercial banks



Note: *Average during 2010-2016, excluding periods with government's stimulus measures (November 2015-April 2016)

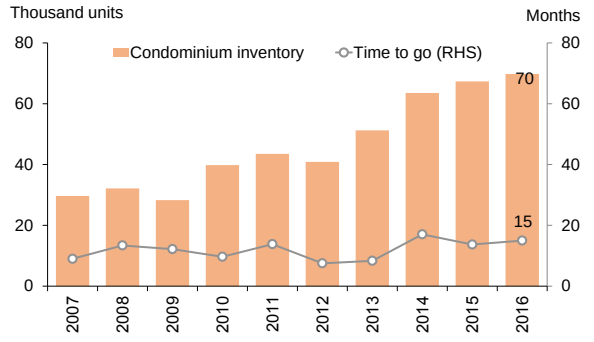
Source: Bank of Thailand

New residential projects launched in Bangkok and its vicinity



Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand

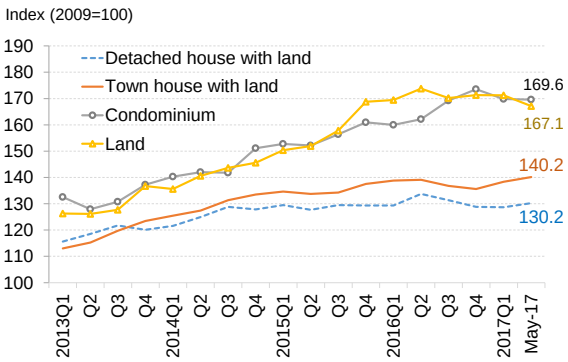
Condominium inventory in Bangkok and vicinity and 'Time to go'



Note: 'Time to go' is the time length that all housing inventory will be sold out, using the average sale rate in the previous 12 months.

Source: AREA and calculation by Bank of Thailand

Real estate price indices

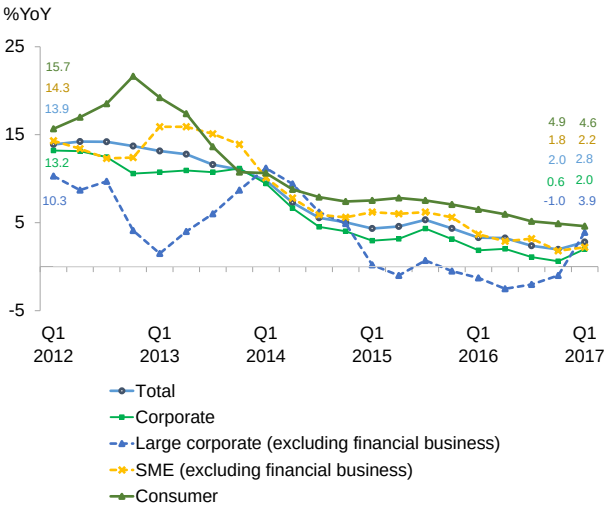


Source: Bank of Thailand

Stability: financial institutions

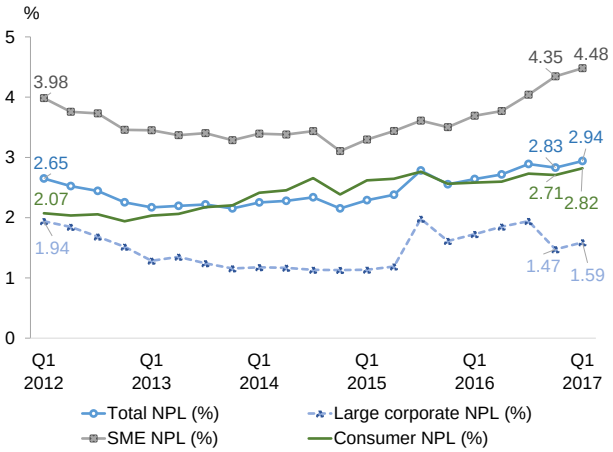
Credit growth slightly increased in the first quarter mainly due to loans to large corporates. While loan quality continued to deteriorate, the financial system remained sound with high levels of provisions and capital buffers.

Credit growth in the commercial bank system



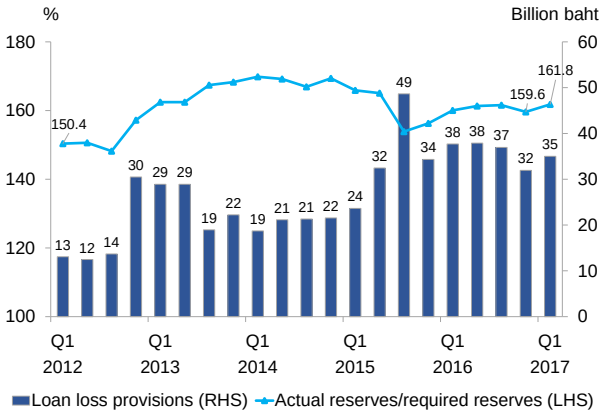
Source: Bank of Thailand

Non-performing loan (NPL)



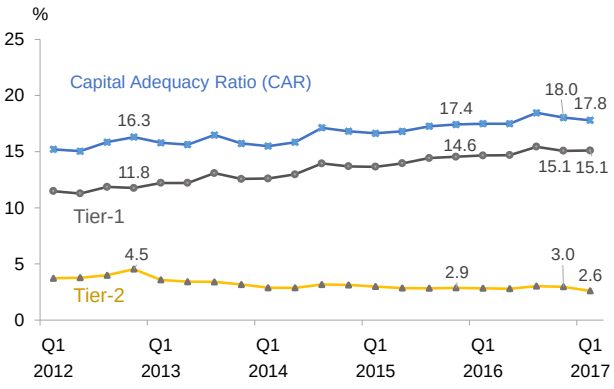
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

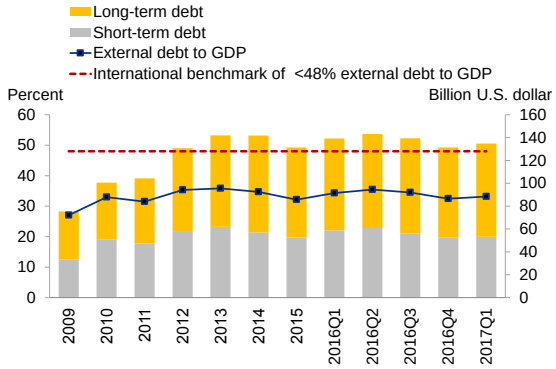


Source: Bank of Thailand

Stability: external position

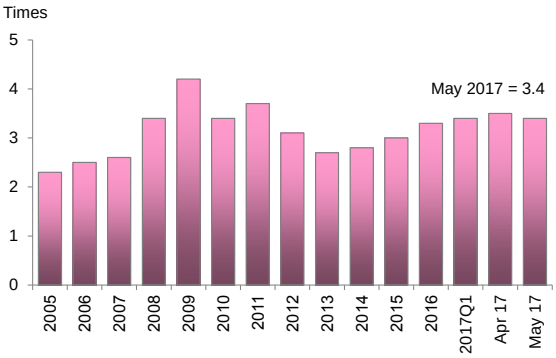
Thailand's external stability remained strong owing to a low level of external debt and international reserves were at a high level relative to short-term debt.

Thailand's external debt



Source: Bank of Thailand

Reserve to short-term debt

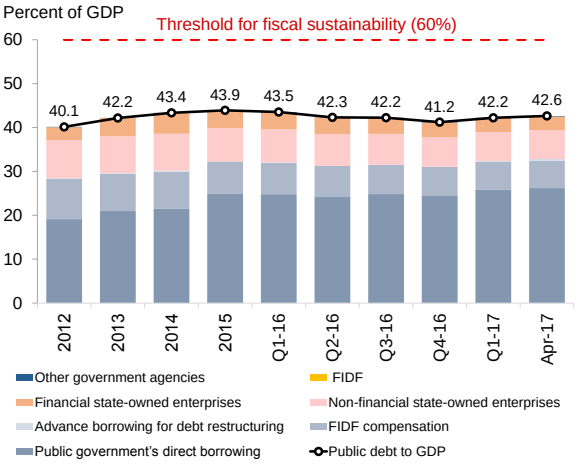


Source: Bank of Thailand

Stability: the fiscal sector

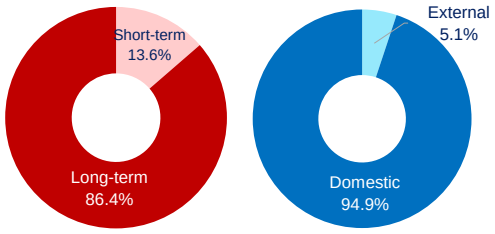
Fiscal stability remained sound. The ratio of public debt to GDP stayed below the fiscal sustainability threshold.

Public debt to GDP



Note: Calculated by GDP with Chain Volume Measure
Source: Public Debt Management Office

Outstanding debt as of April 2017



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office

Pursuing Sustainable
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